



CV2 Type Oil-Immersed Vacuum On-Load Tap-Changer

Technical Data

HM 0.154.5601-08.15/2024

Caution:
For safe and proper use,
follow these instructions.
Retain for future reference.



TABLE OF CONTENTS

1. General	01
2. Technical Specifications	02
3. Model Designation	03
3.1 Model Designation Structure	03
3.2 Tap Selector Basic Connection Types	03
3.3 Basic Wiring Diagram	03
4. Technical Performance Parameters	05
4.1 Rated Step Capacity (PStn)	05
4.2 Breaking Capacity	05
4.3 Short-Circuit Withstand Capability	05
4.4 Operating Conditions	06
4.5 Voltage Gradient Across Components of the Selector Switch	06
4.6 Internal Insulation Level	07
4.7 Insulation Level to Ground	08
4.8 Installation Method	08
5. Special Designs	09
6. Motor Drive Unit	10
7. Tap Changer Controller	11
7.1 SHM-K Remote Digital Controller	11
7.2 HMC-3C On-Load Tap Changer Position Indicator	11
8. Tap Changer Accessories	12
8.1 Bevel Gearbox	12
8.2 Protective Relay	12
9. Appendix	13

1. General

The CV2 type oil-immersed vacuum on-load tap-changer (hereinafter referred to as "OLTC") is an improved vacuum composite product, suitable for oil-immersed transformers with a maximum system voltage of 40.5 kV to 145 kV, a maximum rated through-current of 600 A, and a rated frequency of 50 Hz or 60 Hz. It complies with the requirements of GB/T 10230.1-2019 and IEC 60214-1:2014, "Tap-changers – Part 1: Performance requirements and test methods."

The OLTC is mounted on the transformer tank cover via a mounting flange.

The OLTC is driven by a motor drive unit (MDU). It connects to the MDU via a Top gearbox, a Driving shaft, and a Bevel gearbox, enabling manual or electric tap-changing operations, either locally or remotely.

The OLTC is available in three connection types: three-phase wye (star) neutral, three-phase delta, and single-phase. The basic connection diagrams are shown in Figure 1-1:

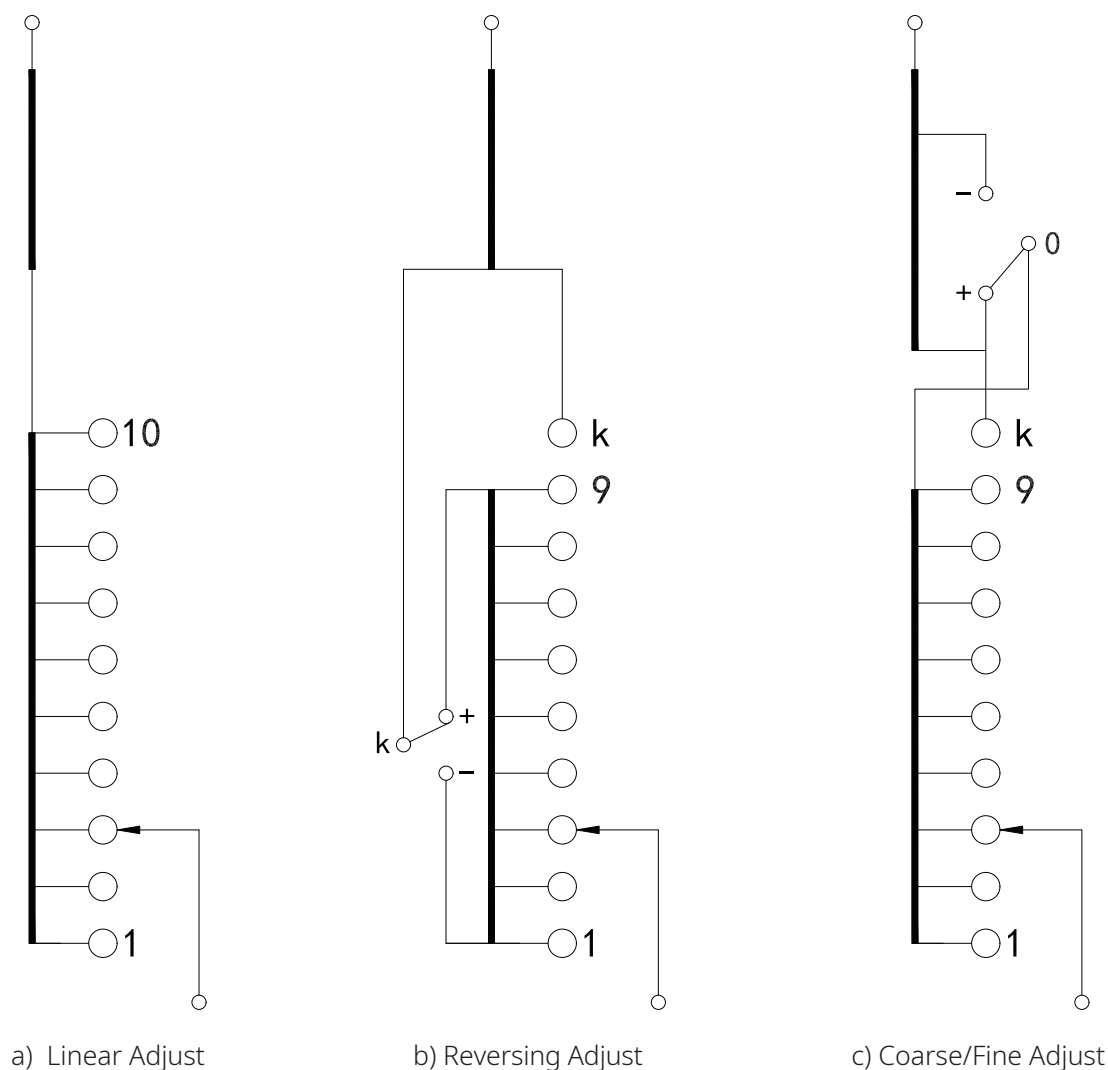


Figure 1-1 Basic Winding Connection for Voltage Regulation

2. Technical Specifications

The OLTC conforms to IEC 60214-1:2014 and GB/T 10230.1-2019 standards. Its technical parameters are listed in Table 2-1.

Table 2-1 Technical Data for On-Load Tap-Changers

Type		CV2 III 350			CV2 III 600		
Max. Rated Through Current (A)		350			600		
Phase		I	III	III	I	III	III
Connection		-	Y	D	-	Y	D
Max. Rated Step Voltage (V)		2000 for 10 contacts 1500 for 12 contact			2000 for 10 contacts 1500 for 12 contact		
Rated Step Capacity (kVA)		700			800		
Short circuit capacity (kA)	Thermal(3s)	5			8		
	Dynamic (peak)	12.5			20		
Max. operating positions		12 without change-over selector; 23 with change-over selector			12 without change-over selector; 23 with change-over selector		
Insulation level to earth (kV)	The highest voltage for equipment (kV)	40.5	72.5		126	145	
	Rated separate source AC withstand voltage (kV/50Hz, 1min)	95	140		230	275	
	Rated lightning impulse withstand voltage- full wave (1.2/50μs)	250	350		550	650	
	Rated lightning impulse withstand voltage-chop wave (3~6μs)	275	285		605	715	
Mechanical life		1,500,000					
Maintenance free		300,000					
Oil compartment	Service pressure	0.03MPa					
	Leakage test	0.1MPa 24h No leakage					
	Over pressure protection	Rupture disc bursts at 0.3±20%MPa					
	Protective relay	Set oil flow speed at 1.0m/s±10%					
Motor drive unit		SHM or CMA7					

3. Model Designation

3.1 Model Designation Structure

OLTC models vary based on the number of phases, maximum rated through-current, maximum system voltage, and connection type. The model designation clearly indicates these parameters. See Figure 3-1 for a detailed explanation of each code.

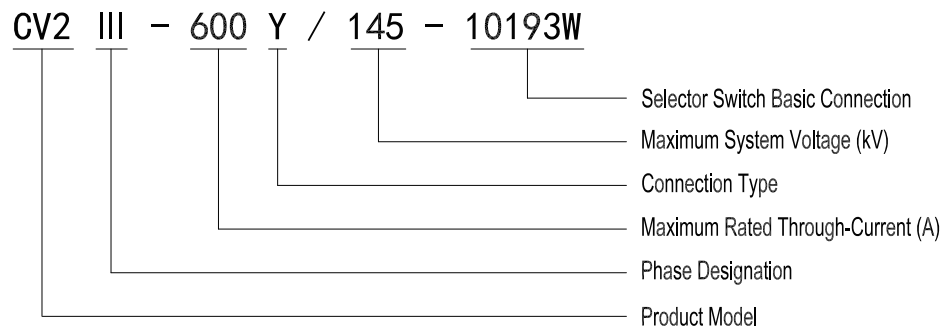


Figure 3-1 OLTC Model Designation Structure

3.2 Selector Switch Basic Connection Types

Selector switches are available in various specifications, depending on the voltage regulation range and winding connection of the transformer. The selector switch specification is defined by the number of selector switch contacts, number of Operation position number, number of intermediate positions, and the type of Change-over selector. The representation of each parameter in the selector switch Product model designation is shown in Figure 3-2.

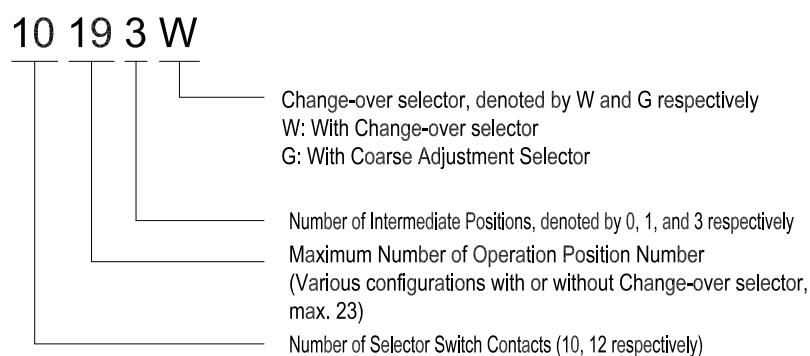


Figure 3-2 Basic Connection of Selector Switch

3.3 Basic Wiring Diagram

Different tapping arrangements of the transformer regulating winding correspond to different basic wiring diagrams for the OLTC. Figure 3-3 shows common basic wiring diagrams. Special designs are available for specific user requirements.

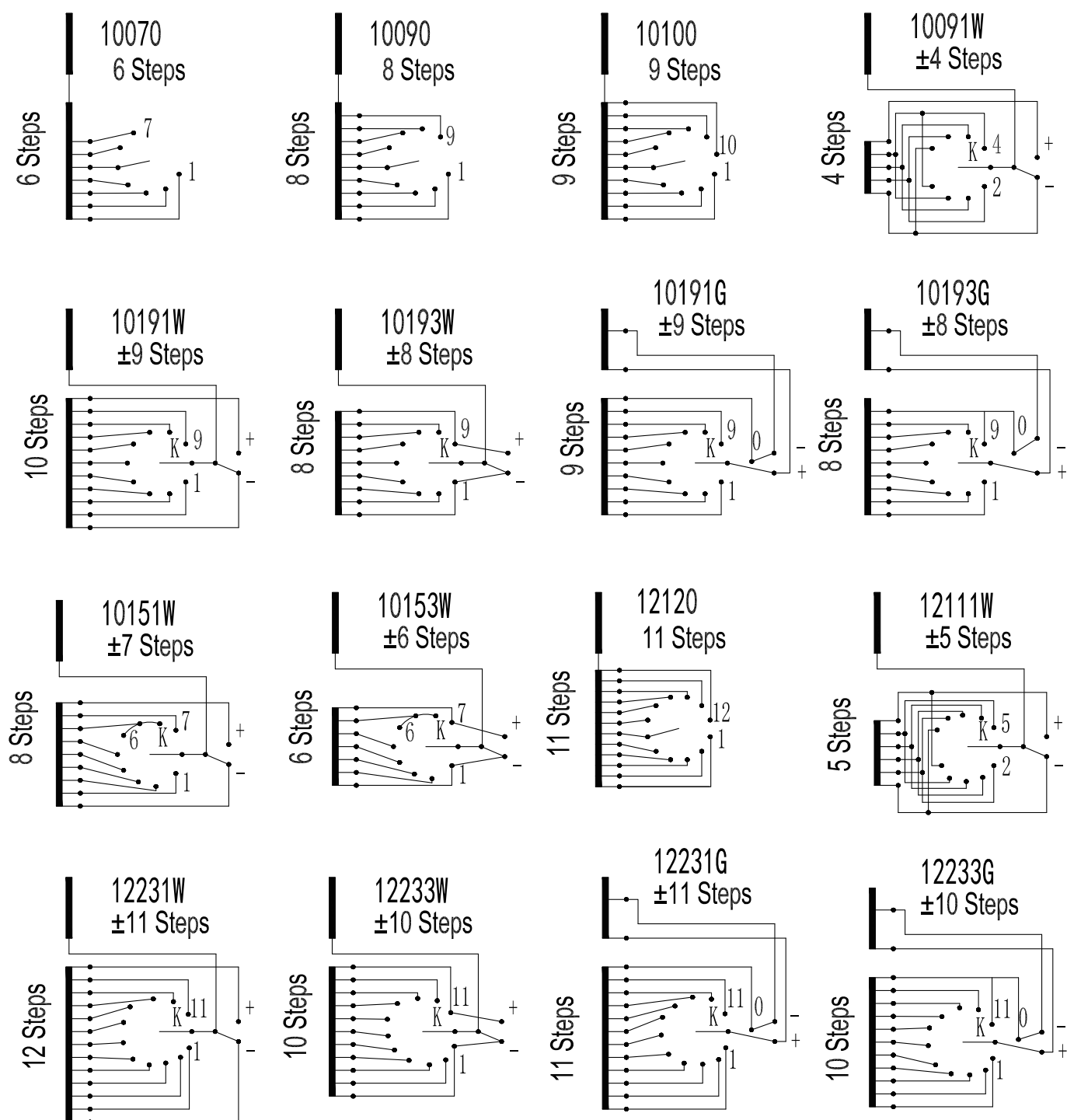


Figure 3-3 Basic Wiring Diagram of On-Load Tap-Changer

4. Technical Performance Parameters

4.1 Rated Step Capacity (P_{Stn})

Step capacity is defined as the product of the step voltage and the load current, i.e., $P_s = U_i I_u$. Rated step capacity is the maximum step capacity of the OLTC under continuous operation, i.e., $P_{Stn} = I_{ux} \times U_i$. See Figure 4-1 for rated step capacities.

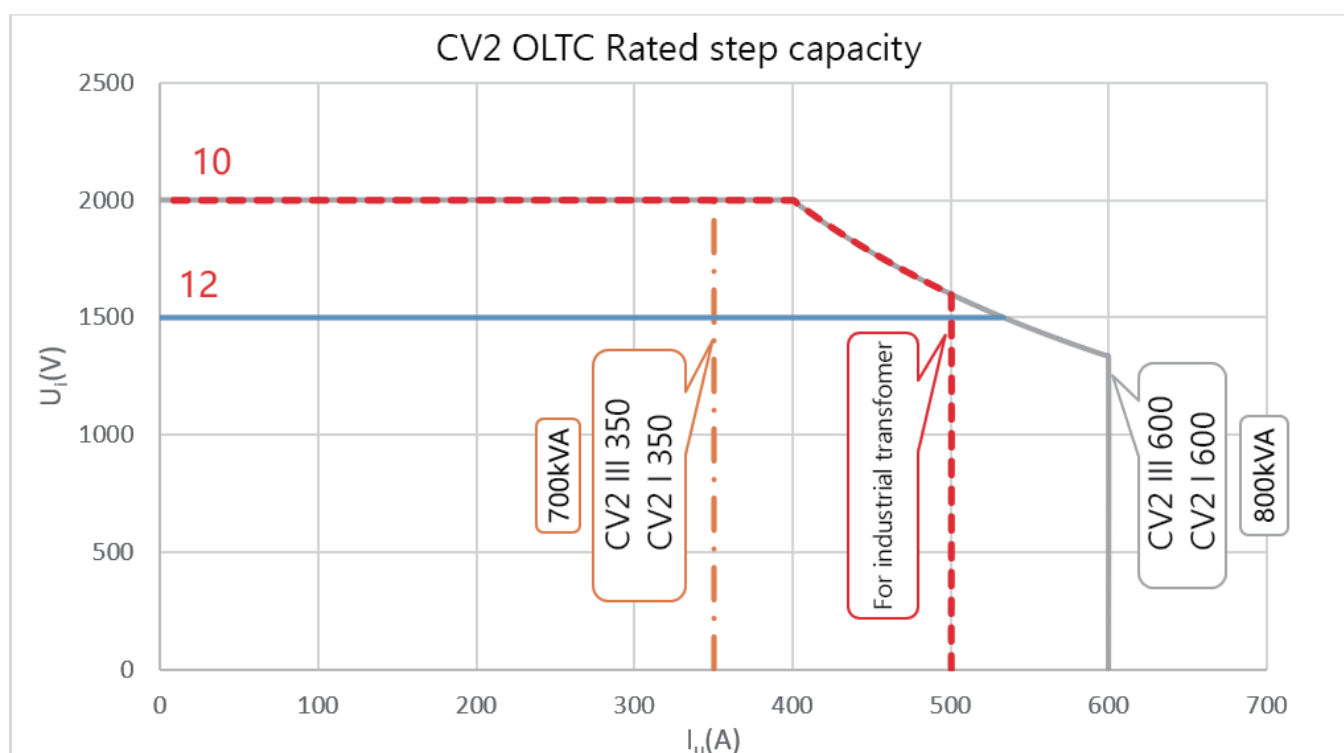


Figure 4-1 Rated Switching Capacity of Three-Phase On-Load Tap-Changer (I_u , U_i)

4.2 Breaking Capacity

Breaking capacity (P_{stmax}) is the maximum switching capacity under load that can be safely interrupted. According to IEC 60214-1:2014, the breaking capacity must be at least twice the rated switching capacity (P_{stn}), i.e., twice the maximum rated through-current (I_{um}) at the relevant rated step voltage (U_i).

4.3 Short-Circuit Withstand Capability

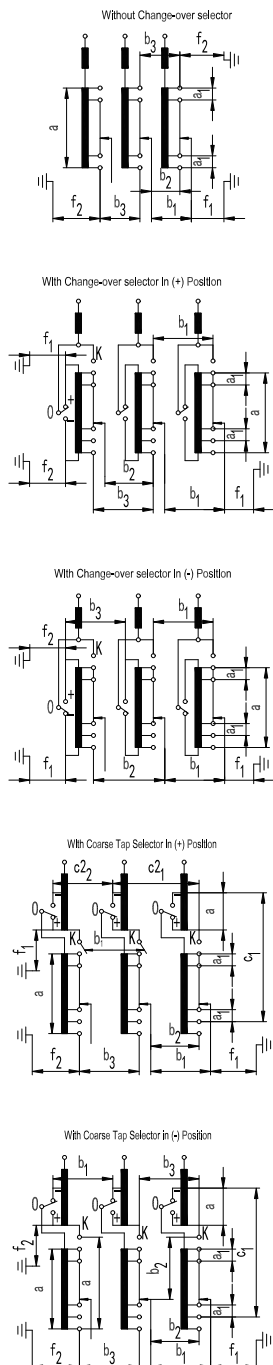
According to IEC 60214-1:2014, all continuously current-carrying contacts of the OLTC must withstand a short-circuit current for 2 s ($\pm 10\%$) without welding, deformation, or mechanical damage. The initial peak current must be 2.5 ($\pm 5\%$) times the RMS value of the rated short-circuit current. Refer the short circuit test current values to Table 1.

4.4 Operating Conditions

1. The operating temperature range for the OLTC in oil is $-25\text{ }^{\circ}\text{C}$ to $+105\text{ }^{\circ}\text{C}$.
2. The ambient temperature range for the OLTC installation location is $-25\text{ }^{\circ}\text{C}$ to $+40\text{ }^{\circ}\text{C}$.
3. After installation on the transformer, the verticality of the OLTC to the ground should not exceed 2%.
4. The installation location must be free of heavy dust and explosive or corrosive gases.

Note: For more severe environmental conditions for the OLTC or MDU, contact Huaming's technical department for confirmation.

4.5 Voltage Gradient Across Components of the Selector Switch



a_1 = Between tap selector contacts (connected or unconnected) at any tap position.

a = Between the start and end of the tap winding. If a coarse tap winding is present, it is also between the start and end of the coarse tap winding.

Note: When the coarse tap is in the Change-over selector (-) position: The permissible withstand voltage value "a" must be observed between the end of the coarse tap winding connected to the tap selector K-contacts and the tap selector contacts located at the end of the same tap winding, when subjected to impulse voltages.

b = Between tap selector contacts of different phases, and also between Change-over selector contacts of different phases. This gap is connected to the start (or end) of the tap winding or to one of the tap selector contacts.

The permissible voltage between the delta-connected tap changer contacts varies with the position of the Change-over selector and the tap selector. Therefore, different withstand voltage values for b_1 , b_2 , and b_3 must be observed for this insulation gap:

b_1 = Between selected contacts of different phases;

b_2 = Between the selected contact of one phase and the unselected contact of another phase;

b_3 = Between unselected contacts of different phases.

f = Between tap selector contacts and Change-over selector (+) contacts to ground.

Two values are used for the delta connection design:

f_1 = Between the terminals and ground when the Change-over selector is in the (+) position, and between the Change-over selector (+) contacts and ground;

f_2 = Between tap selector unselected contacts when the Change-over selector is in the (-) position, and between the Change-over selector (+) contacts and ground.

Additionally, when the coarse tap winding is located in the (+) position of the Change-over selector:

c_1 = Between the (-) contacts of the Change-over selector and the terminals of the same phase.

c_2 = Between the (-) contacts of the Change-over selector of a different phase, and between the (-) contact of the Change-over selector and the (+) contact of the Change-over selector of the other phase.

The following different permissible values are used for the delta-connected switches:

c_{21} = Between the Change-over selector (-) contact and the (+) contact of the Change-over selector of the other phase;

c_{22} = Between the Change-over selector (-) contacts of different phases.

Figure 4-2 Voltage Gradient at Each Location

4.6 Internal Insulation Level

The internal insulation level of the OLTC (see Table 4-1) is related to the basic voltage regulation method (see Figure 4-2). It must be verified during OLTC selection to ensure that the internal insulation level meets the withstand voltage requirements.

Table 4-1 Internal Insulation Level (kV)

Insulation Distance Symbol	Switch Type	Test Method	III 350/600A (D)			III 350/600A (Y)
a	10 Contacts	kV 1.2/50μs	200			
		kV 50Hz 1min	50			
	12 Contacts	kV 1.2/50μs	200			
		kV 50Hz 1min	60			
a1	10 Contacts	kV 1.2/50μs	200			
		kV 50Hz 1min	50			
	12 Contacts	kV 1.2/50μs	200			
		kV 50Hz 1min	50			
b	40.5 kV		b ₁	b ₂	b ₃	
		kV 1.2/50μs	200	250	300	
		kV 50Hz 1min	70	80	90	
	72.5 kV	kV 1.2/50μs	350	490	520	
		kV 50Hz 1min	140	165	180	
	126 kV	kV 1.2/50μs	550	570	600	
		kV 50Hz 1min	230	240	250	
	145 kV	kV 1.2/50μs	650	730	800	
		kV 50Hz 1min	275	285	300	
C1	40.5 kV 72.5 kV	kV 1.2/50μs	400			400
		kV 50Hz 1min	100			100
	126 kV	kV 1.2/50μs	470			470
		kV 50Hz 1min	120			120
	145 kV	kV 1.2/50μs	500			500
		kV 50Hz 1min	130			130
C2	40.5 kV		C2 ₁	C2 ₂		
		kV 1.2/50μs	250	300	200	
		kV 50Hz 1min	80	90	85	
	72.5 kV	kV 1.2/50μs	490	520	200	
		kV 50Hz 1min	165	180	85	
	126 kV	kV 1.2/50μs	570	600	200	
		kV 50Hz 1min	240	250	85	
	145 kV	kV 1.2/50μs	730	800	200	
		kV 50Hz 1min	285	300	85	

Insulation Distance Symbol	Switch Type	Test Method	III 350/600A (D)		III 350/600A (Y)
f	40.5 kV		f_1	f_2	
		kV 1.2/50 μ s	200	250	200
	72.5 kV	kV 50Hz 1min	85	85	85
		kV 1.2/50 μ s	350	490	350
	126 kV	kV 50Hz 1min	140	165	140
		kV 1.2/50 μ s	550	570	550
	145 kV	kV 50Hz 1min	230	240	230
		kV 1.2/50 μ s	650	730	650
		kV 50Hz 1min	275	285	275

4.7 Insulation Level to Ground

The insulation level to ground is defined as the insulation between the live parts of the OLTC and grounded components. It is determined by the AC power-frequency one-minute voltage withstand test and the impulse voltage withstand test. It is related to the location of the transformer tap winding, the regulation range and method, the winding connection method and arrangement, and the rated voltage of the transformer windings. It is determined by the insulation level to ground of the transformer regulating winding.

The insulation level to ground of the OLTC must be selected from the standard values specified in IEC 60214-1:2014 and GB/T 10230.1-2019 (see Table 4-2) based on the maximum operating voltage (U_m) of the equipment, using the lowest possible insulation value to meet the entire operating range.

Table 4-2 Insulation Level to Ground

Maximum System Voltage (U_m)	Rated Applied Withstand Voltage (50 Hz, 1 min)	Rated Lightning Impulse Withstand Voltage (Full Wave, 1.2/50 μ s)
40.5	95	250
72.5	140	350
126	230	550
145	275	650

4.8 Installation Method

The OLTC is mounted on the transformer tank cover using a mounting flange. The transformer must be equipped with a mounting flange. See Appendix Figures 9 and 10 for dimensions. For bell-type transformers, the OLTC support flange serves only as temporary support during transformer wiring. After the transformer is installed, the OLTC is secured to the transformer mounting flange.

5. Special Designs

Potential Connection of Regulating Winding

In on-load tap-changing transformers with high voltage ratings and large regulating ranges, the regulating winding is momentarily electrically disconnected from the main winding during Change-over selector operation, entering a "floating" state. At this time, the regulating winding will acquire a new potential, determined by the coupling capacitance to ground (C_e) and the coupling capacitance to adjacent windings (C_w) (see Figure 5-2). This potential is typically different from the regulating winding potential before the Change-over selector operation. The potential difference between the two is called the offset voltage, which appears as a recovery voltage at the Change-over selector contact break. When the offset voltage reaches a critical value, it can cause a discharge at the Change-over selector, generating a large amount of gas, which is a serious problem. Therefore, if the OLTC offset voltage exceeds 15 kV (RMS), the potential connection of the regulating winding must be considered to prevent discharge during Change-over selector operation.

To facilitate Huaming's determination of the load on the Change-over selector and the size and number of potential Grading ring resistor elements, the following transformer parameters must be specified during OLTC selection:

1. All transformer performance parameters: rated capacity, rated voltage, voltage regulation range, winding connection type, and insulation level;
2. Relative arrangement of windings: the relative position of the regulating winding with respect to adjacent windings or winding components;
3. The voltage at the terminals adjacent to the regulating winding and the electrical position of that coil in the transformer winding;
4. Capacitance between the regulating winding and adjacent winding components;
5. Capacitance between the regulating winding and ground or adjacent grounded winding (if present);
6. Impulse load across half of the regulating winding;
7. Operating and power-frequency test voltages across half of the regulating winding.

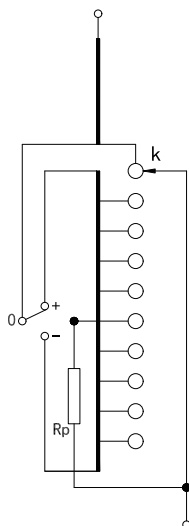


Figure 5-1 Tie-in Resistor Fixed Connection

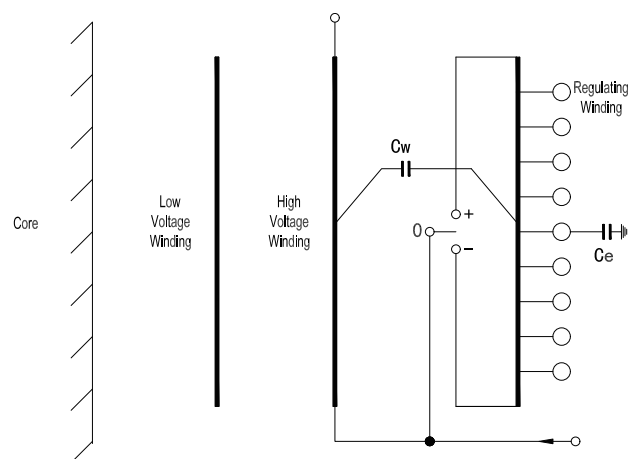


Figure 5-2 Arrangement of Reversing Winding in a Two-Winding Transformer

6. Motor Drive Unit

The OLTC is driven by a Motor Drive Unit (MDU). Users can select the appropriate MDU based on the OLTC model. Technical parameters for the MDUs are listed in Table 6-1.

Table 6-1 Motor Drive Unit Technical Parameters (Partial)

Motor Drive Unit		SHM-D	CMA7	
Parameter				
Motor	Rated Power (W)	750	750	1100
	Rated Voltage (V)	200~240 (AC)	Three-Phase 380 (AC)	
	Rated Current (A)	4	2.0	2.8
	Frequency (Hz)	50 or 60	50 or 60	
	Speed (r/min)	330 Direct Drive	1400	
Output Shaft Torque (Nm)		35 Static Torque	18	26
Rev. of drive shaft/ per switching operation		33	33	
Handle Revolutions per Tap Change		33	33	
Motorized Operating Time per Tap Change (s)		Approx. 6	Approx. 5	
Maximum Number of Operating Positions		107	107	
Control Circuit and Heater Voltage (V)		220 (AC)	220 (AC)	
Heater Power Consumption (W)		100	50	
Auxiliary Circuit Insulation Test (kV/50 Hz, 1 min)		2	2	
Weight (kg)		80	90	
Ingress protection		IP66	IP56	
Mechanical Endurance of MDU		≥2,000,000	≥800,000	

7. Tap Changer Controller

7.1 SHM-K Remote Digital Controller

The SHM-K remote digital controller (hereinafter referred to as SHM-K) can be used with the MDU for manual and automatic control of the OLTC. The SHM-K features a color, high-resolution touch screen with a 400 MHz main processor, 256 MB Flash memory, and 64 MB RAM, providing fast operation and large memory capacity. It communicates bidirectionally and digitally with the local control module via fiber optic cable.

SHM-K Remote Digital Controller Technical Parameters:

Operating Voltage: 85 V to 264 V (AC), 50/60 Hz; 100 V to 250 V (DC)
Power Consumption: Approx. 7 W
Maximum Number of Display position : 107
Operating Environment: Indoor, 0 °C to 50 °C; Relative Humidity < 95% non-condensing

7.2 HMC-3C On-Load Tap Changer Position Indicator

The HMC-3C on-load tap changer position indicator can be used with the MDU's remote position indication circuit to remotely display the tap position. It also provides "1-N", "Stop", and "N-1" functions, outputs BCD code position signals, and includes a remote control indicator.

HMC-3C Monitor Technical Parameters:

Operating Voltage: 220 V (AC)	Operating Frequency: 50 Hz
Maximum Number of Display position: 107	Operating Temperature: -10 °C to +40 °C

Note: Specify any other power supply technical requirements for the controller at the time of order.

8. Tap Changer Accessories

8.1 Bevel Gearbox

The bevel gearbox connects the horizontal shaft of the OLTC to the vertical shaft of the MDU, transmitting the drive torque to the OLTC. See Appendix Figure 25 for the external dimensions of the bevel gearbox.

8.2 Protective Relay

The Protective relay is a protective device used in oil-immersed OLTCs. In the event of gas generation due to oil decomposition caused by an internal fault in the OLTC, or a surge in oil flow, the relay contacts actuate, connecting the designated control circuit and sending a signal or automatically tripping the transformer.

The Protective relay is installed in the connection pipe between the elbow at the top of the OLTC and the oil conservator, with the arrow on the relay pointing towards the oil conservator. Huaming offers various models of Protective relays. See Appendix Figure 26 for the mounting dimensions of the Protective relay.

9. Appendix

Figure 1 Outline Dimensions of CV2 Type III 350/600 - 10070 (Tank Top) OLTC	14
Figure 2 Outline Dimensions of CV2 Type III 350/600 - 10070 (Bell-Type) OLTC	15
Figure 3 Outline Dimensions of CV2 Type III OLTC with Change-over Selector	16
Figure 4 Outline Dimensions of CV2 Type III OLTC without Change-over Selector	17
Figure 5 Outline Dimensions of CV2 Type I OLTC with Change-over Selector	18
Figure 6 Outline Dimensions of CV2 Type I OLTC without Change-over Selector	19
Figure 7 Outline Dimensions of CV2 Type III 350/600 - 10071 (10091) OLTC	20
Figure 8 Outline Dimensions of CV2 Type III 350/600 - 12111 OLTC	21
Figure 9 Mounting Dimensions of CV2 Type Transformer Mounting Flange (Tank Top) ...	22
Figure 10 Mounting Dimensions of CV2 Type Transformer Mounting Flange (Bell-Type) ..	23
Figure 11 Mounting Dimensions of CV2 Type OLTC (Bell-Type)	24
Figure 12 Mounting Dimensions of Tie-in Resistor	25
Figure 13 Top Cover Arrangement of CV2 Type OLTC.....	26
Figure 14 Basic Connection Diagram of CV2 Type OLTC 10070	27
Figure 15 Basic Connection Diagram of CV2 Type OLTC 10071W	28
Figure 16 Basic Connection Diagram of CV2 Type OLTC 10191W	29
Figure 17 Basic Connection Diagram of CV2 Type OLTC 10191G	30
Figure 18 Basic Connection Diagram of CV2 Type OLTC 10193W	31
Figure 19 Basic Connection Diagram of CV2 Type OLTC 10193G	32
Figure 20 Basic Connection Diagram of CV2 Type OLTC 12231W	33
Figure 21 Basic Connection Diagram of CV2 Type OLTC 12231G	34
Figure 22 Basic Connection Diagram of CV2 Type OLTC 12233W	35
Figure 23 Basic Connection Diagram of CV2 Type OLTC 12233G	36
Figure 24 Mechanical Interlocking Arrangement of Three OLTCs	37
Figure 25 Outline Dimensions of Bevel Gearbox	38
Figure 26 Outline Dimensions of Protective relay	39

Figure 1 Outline Dimensions of CV2 Type III 350/600 - 10070 (Tank Top) OLTC

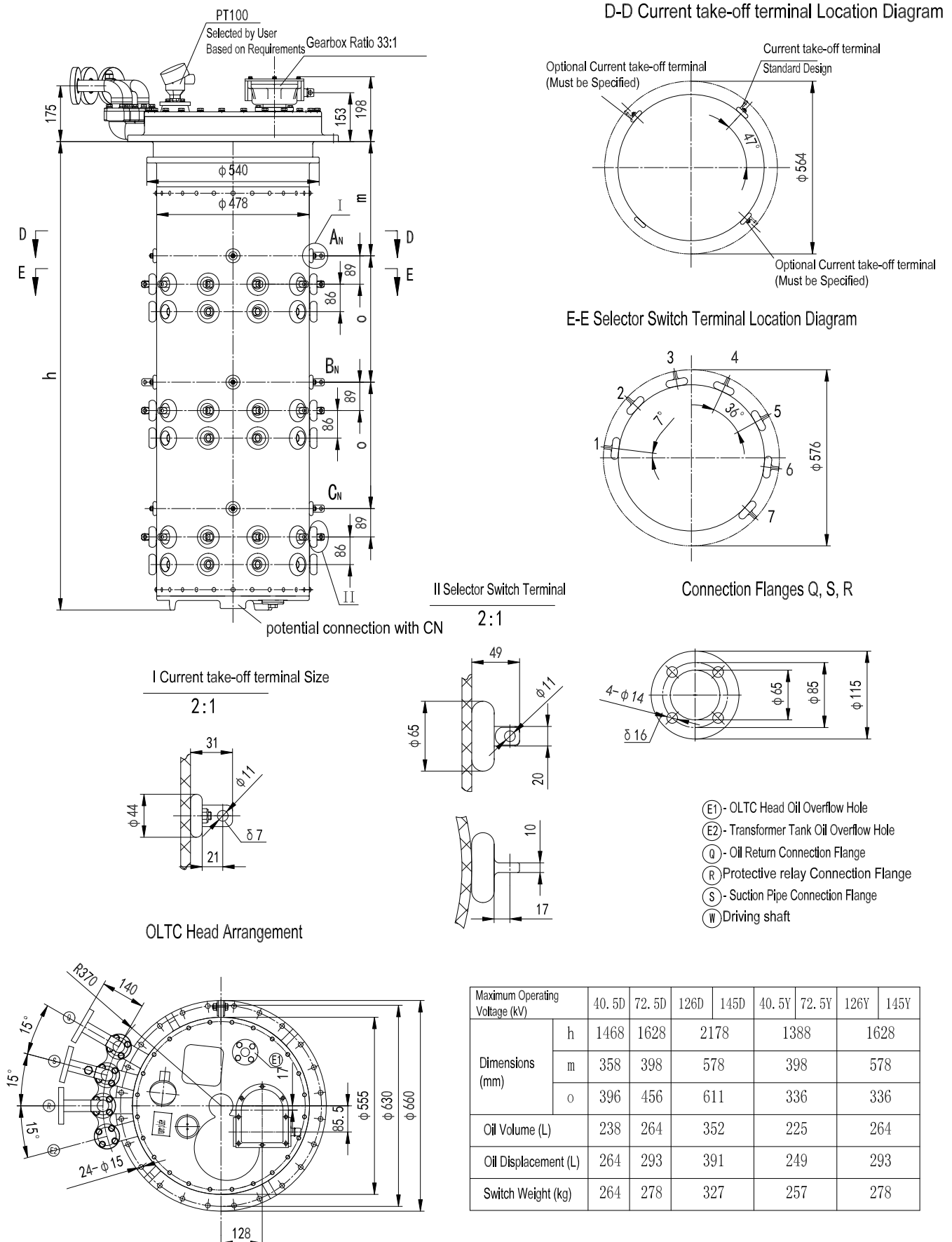


Figure 2 Outline Dimensions of CV2 Type III 350/600 - 10070 (Bell-Type) OLTC

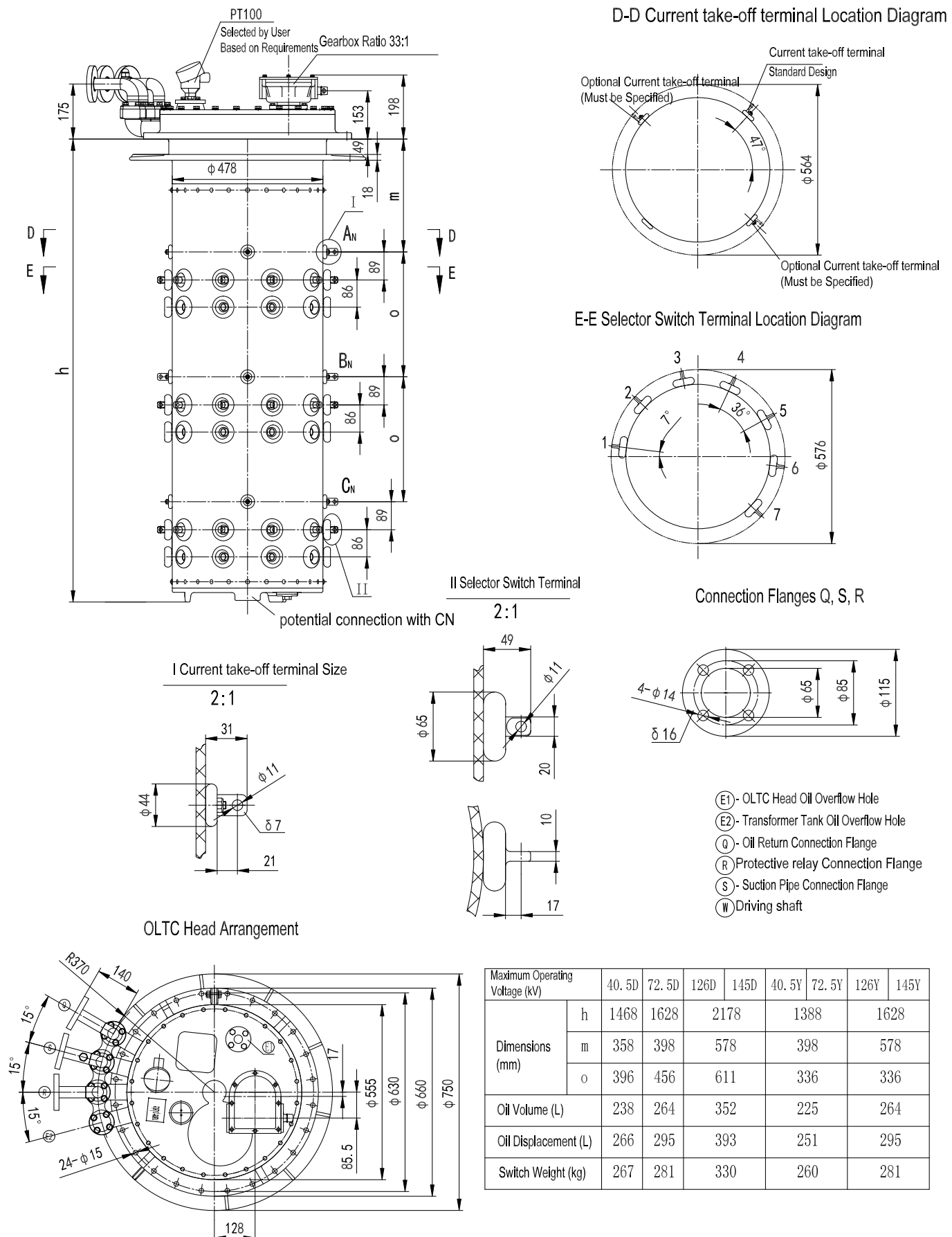
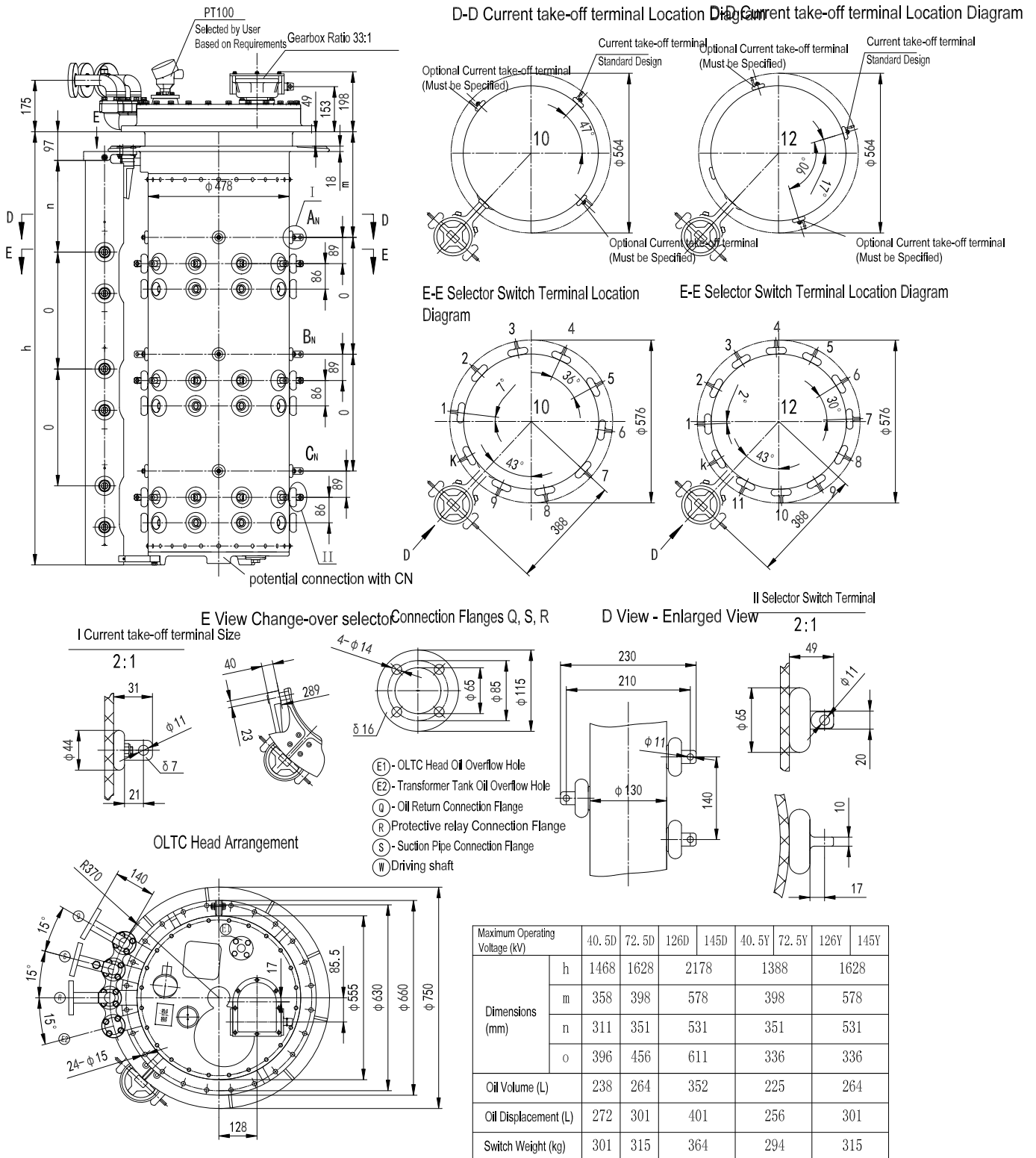


Figure 3 Outline Dimensions of CV2 Type III OLTC with Change-over Selector



100



Figure 5 Outline Dimensions of CV2 Type I OLTC with Change-over Selector

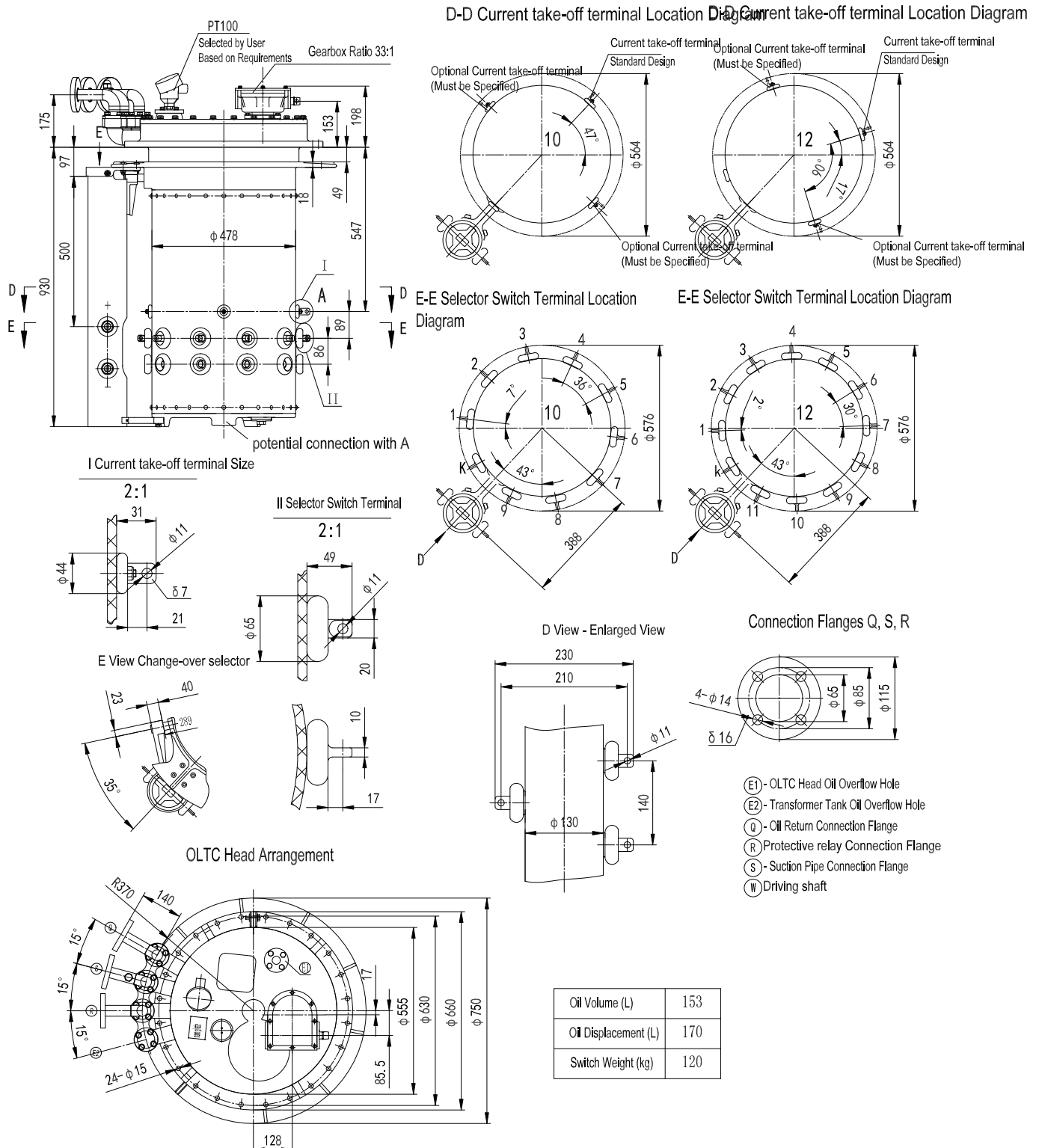


Figure 6 Outline Dimensions of CV2 Type I OLTC without Change-over Selector

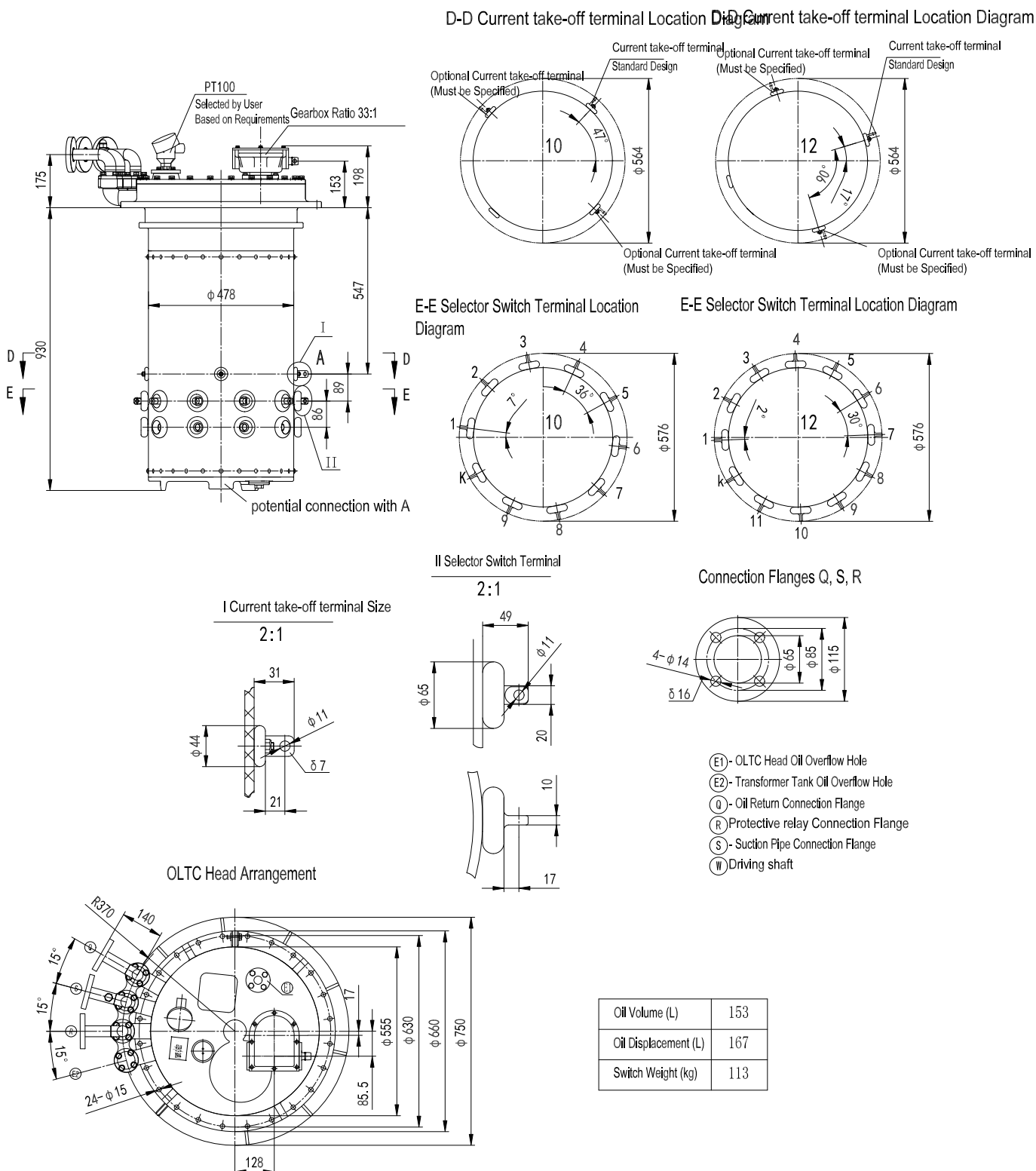


Figure 7 Outline Dimensions of CV2 Type III 350/600 - 10071 (10091) OLTC

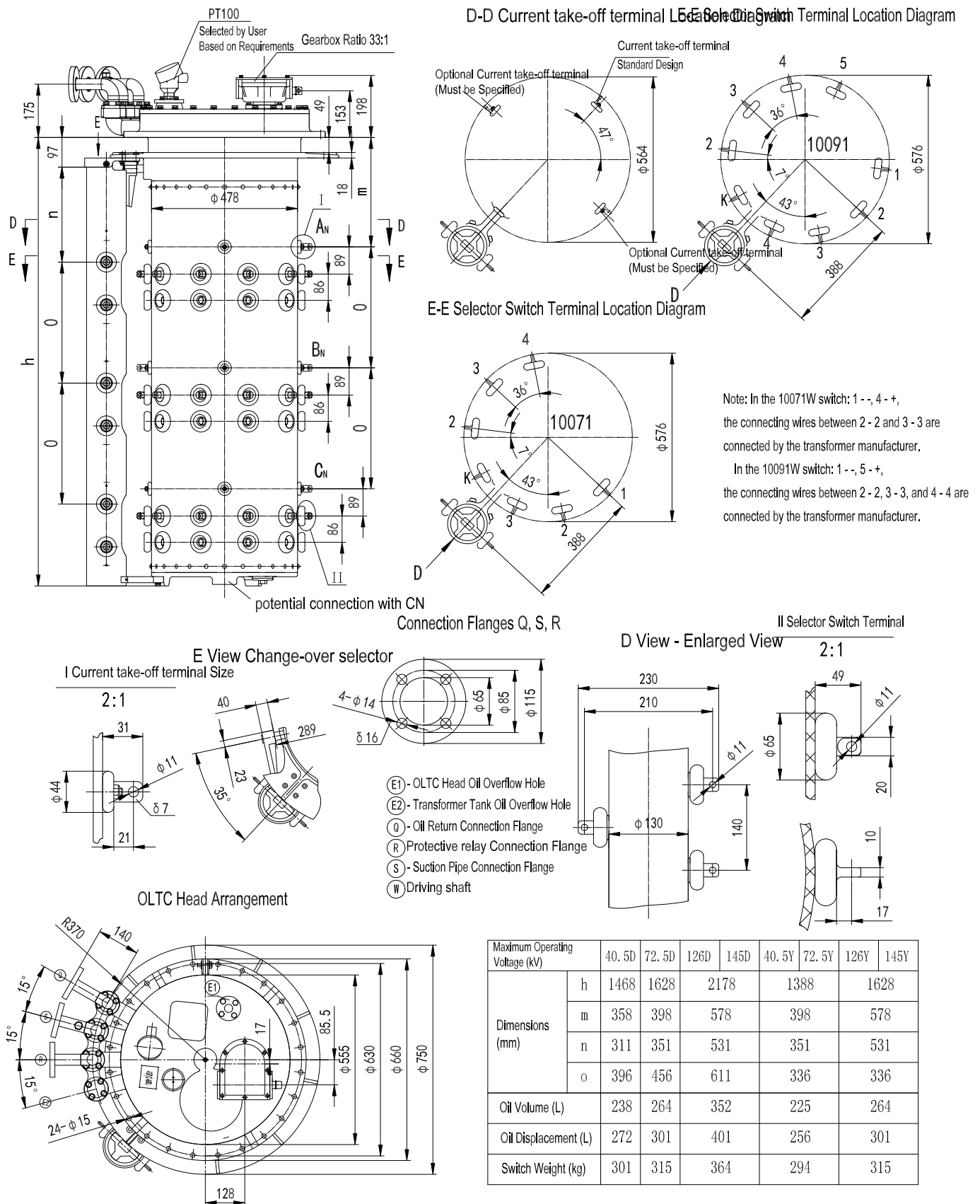






Figure 9 Mounting Dimensions of CV2 Type Transformer Mounting Flange (Tank Top)

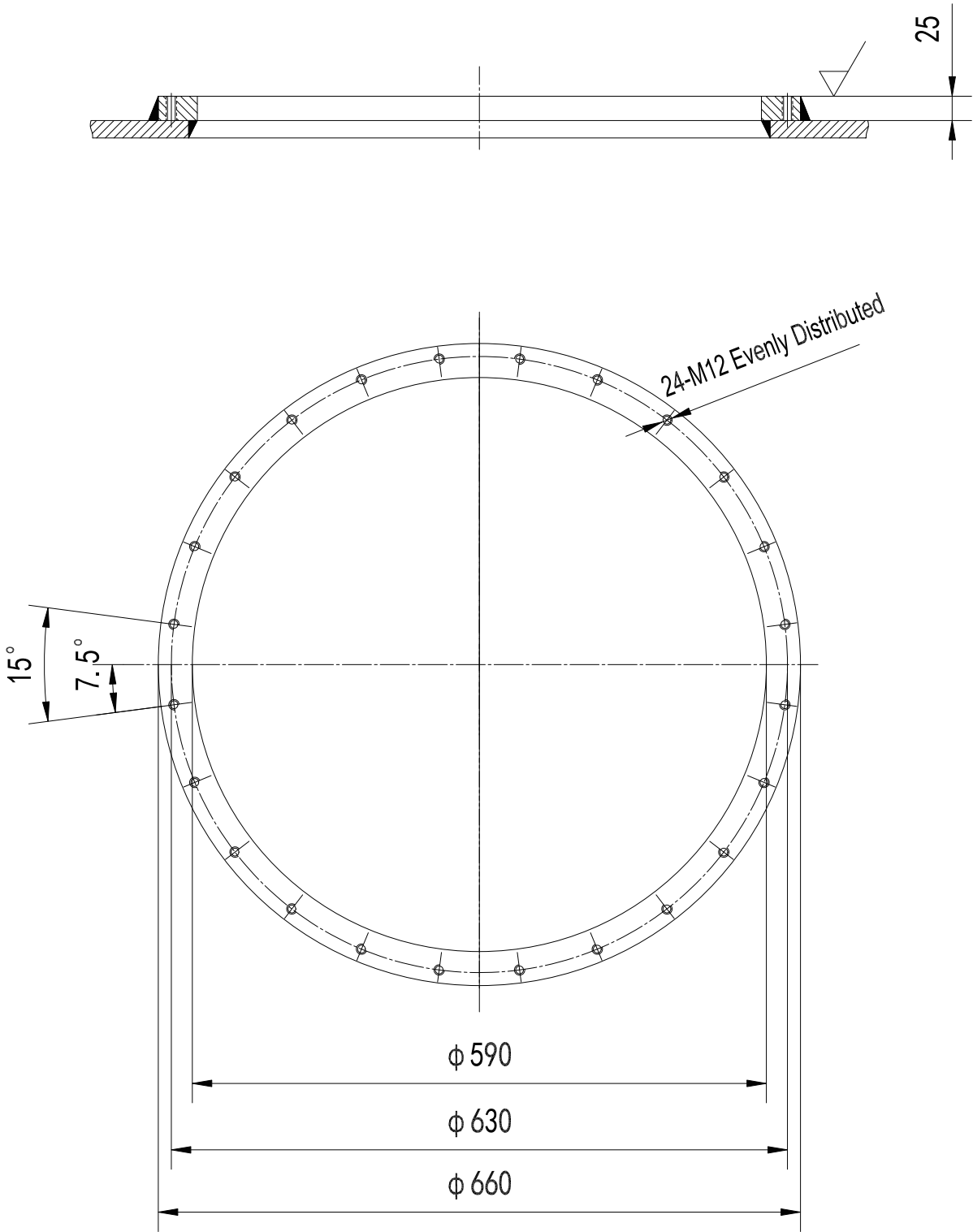


Figure 10 Mounting Dimensions of CV2 Type Transformer Mounting Flange (Bell-Type)

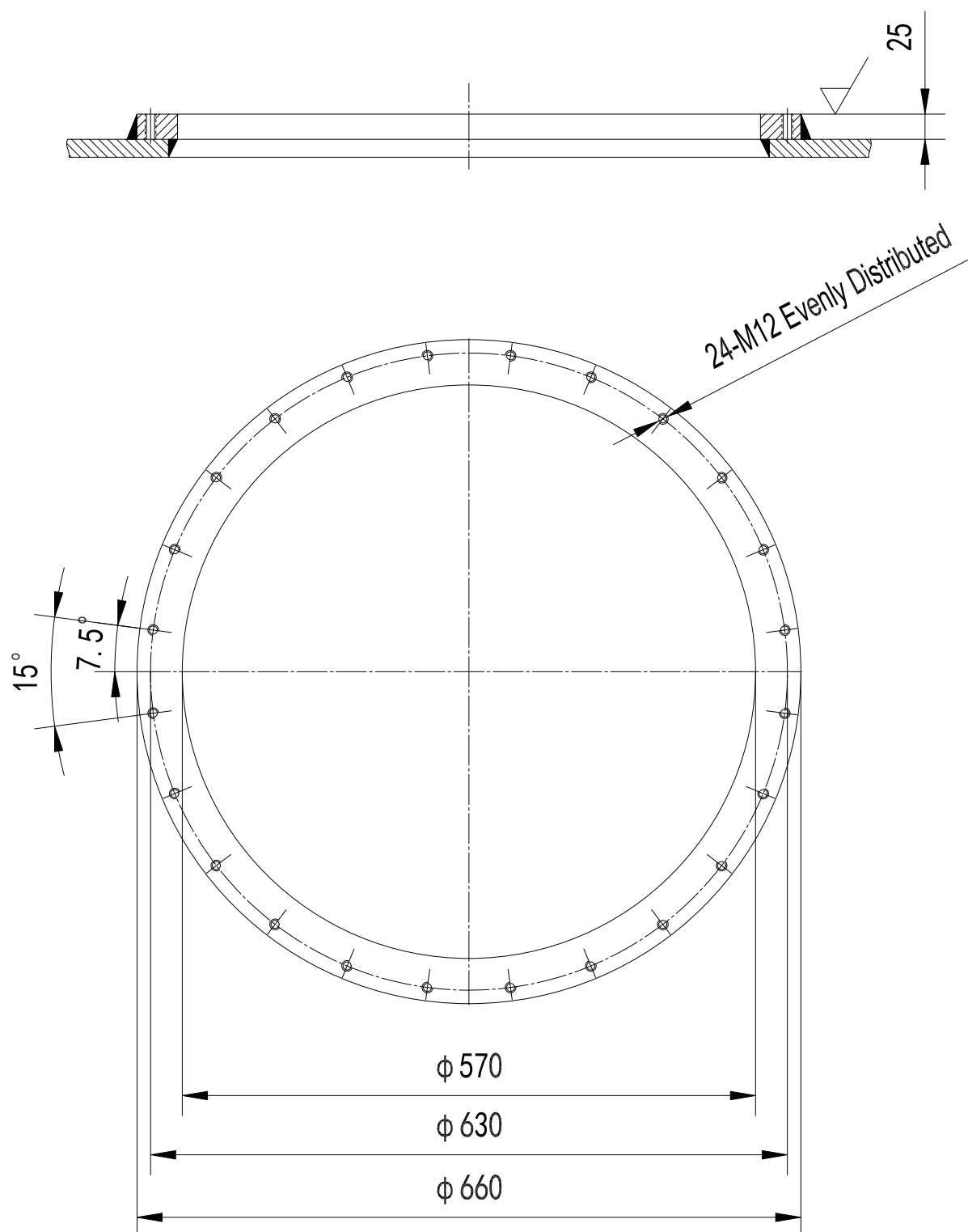




Figure 11 Mounting Dimensions of CV2 Type OLTC (Bell-Type)

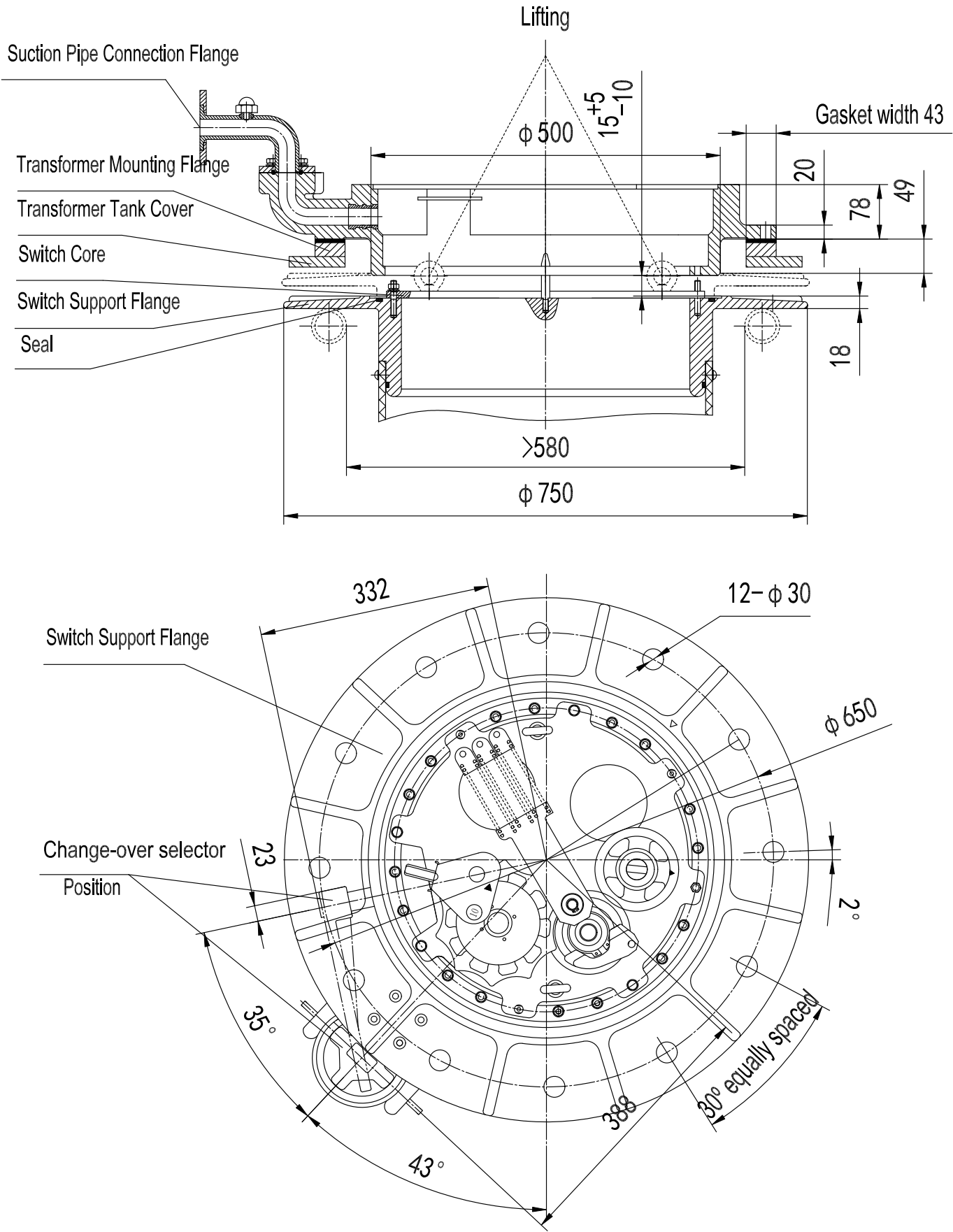
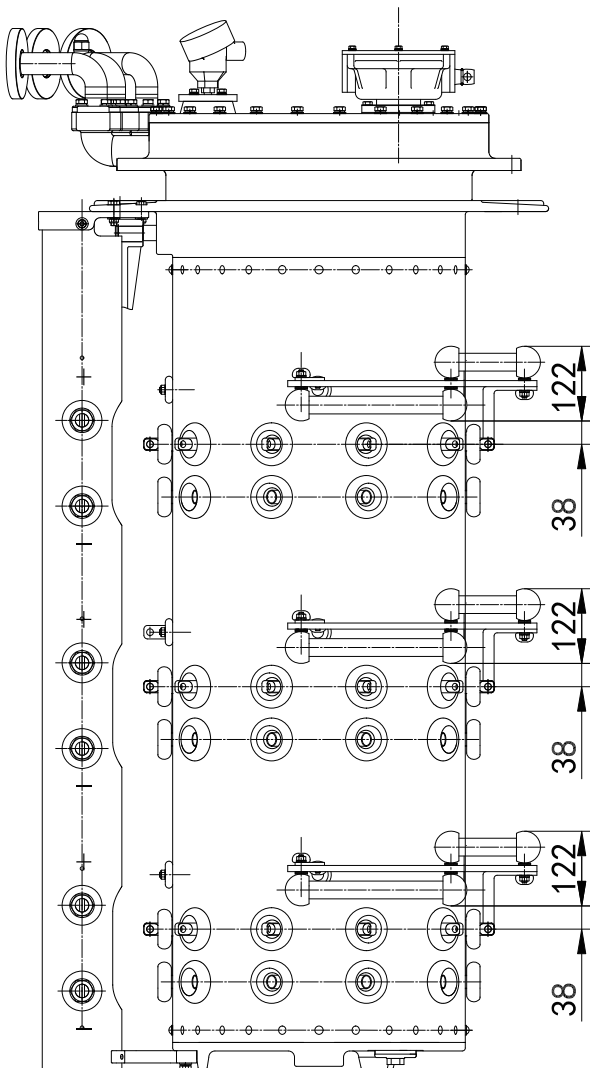


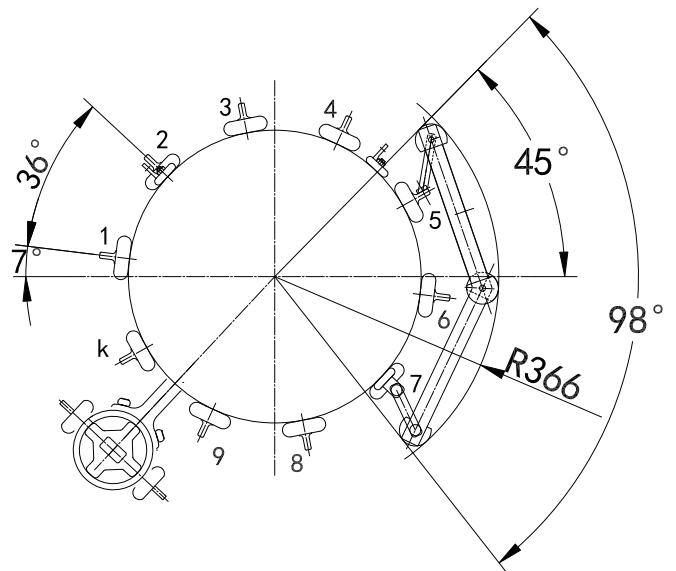
Figure 12 Mounting Dimensions of Tie-in Resistor

When the tap changer selector is in operation, the tap winding is in a "floating" state. Due to the coupling capacitance C_w between the main winding and the tap winding, and a coupling capacitance C_e between the tap winding and the tank, the Change-over selector generates a spark discharge. To reduce spark discharge gas, fixed Tie-in resistors are employed.

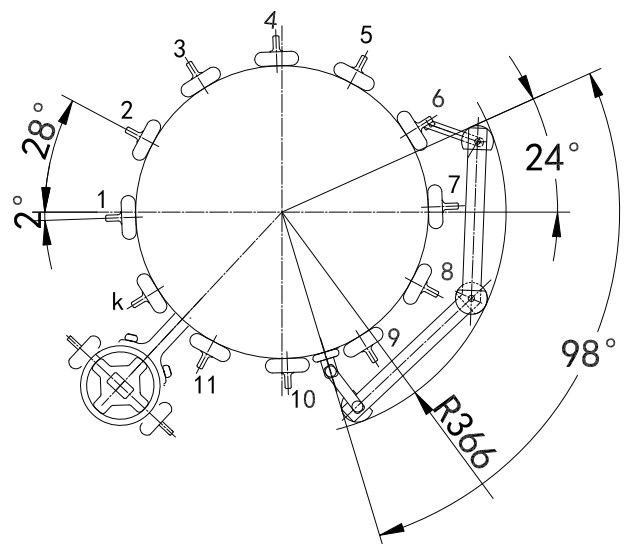
Wall-Mounted Tie-in Resistor Installation Dimensions



10-Contact Installation Dimensions



12-Contact Installation Dimensions



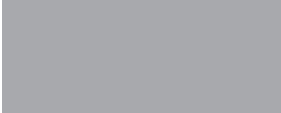
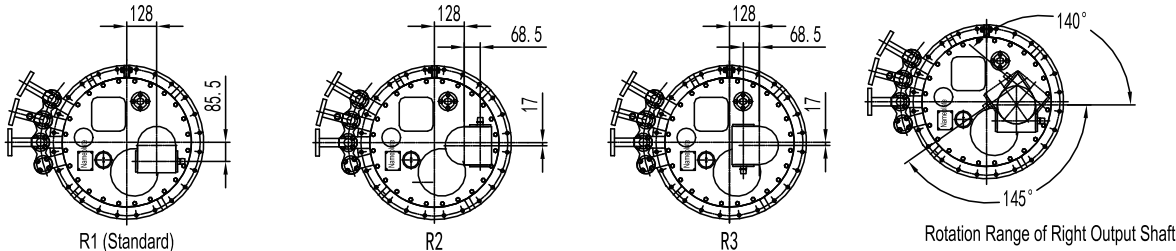


Figure 13 Top Cover Arrangement of CV2 Type OLTC

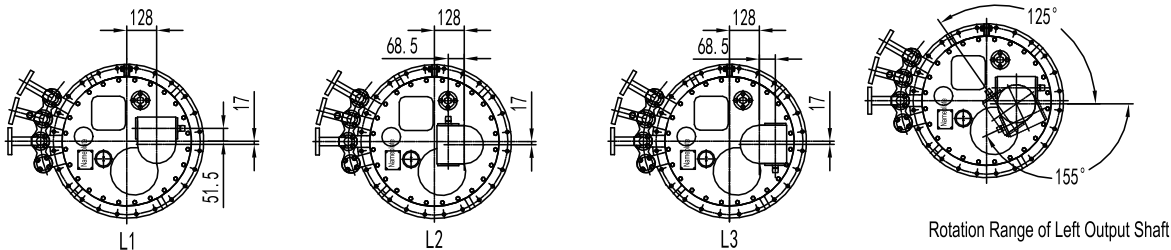
1 Optional Drawing of Top Cover Arrangement without Pressure relief valve/device

When ordering, select the production method for the top cover. Otherwise, it will be produced according to standard drawing R1.
Specify special angle requirements at the time of order.

Right Output Shaft Arrangement:



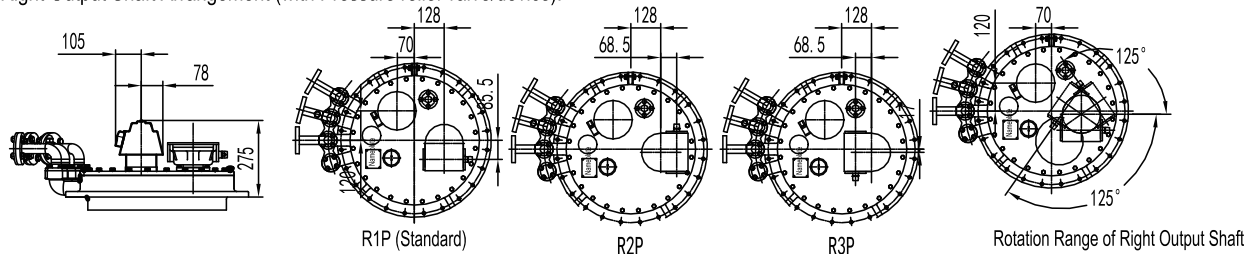
Left Output Shaft Arrangement:



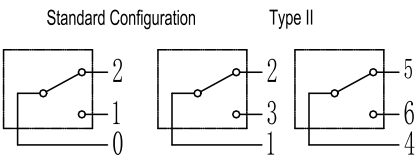
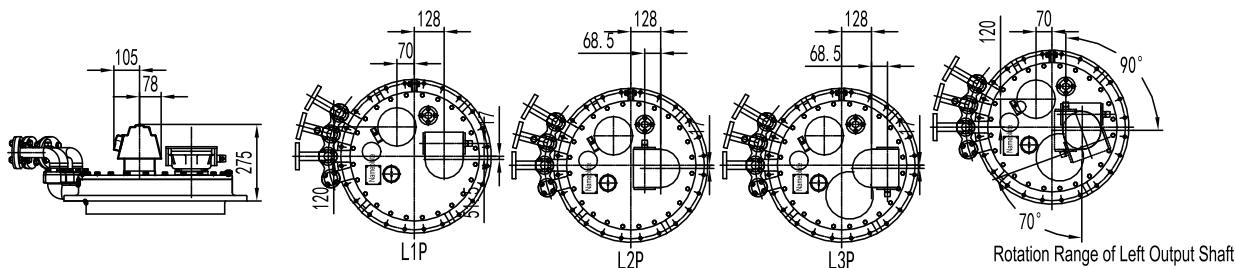
2 Optional Drawing of Top Cover Arrangement with Pressure relief valve/device (Standard pressure relief valve/device diameter is 50 mm)

When ordering the switch, select the top cover arrangement (with Pressure relief valve/device). Otherwise, it will be produced according to standard drawing R1P.
Specify special angle requirements at the time of order.

Right Output Shaft Arrangement (with Pressure relief valve/device):



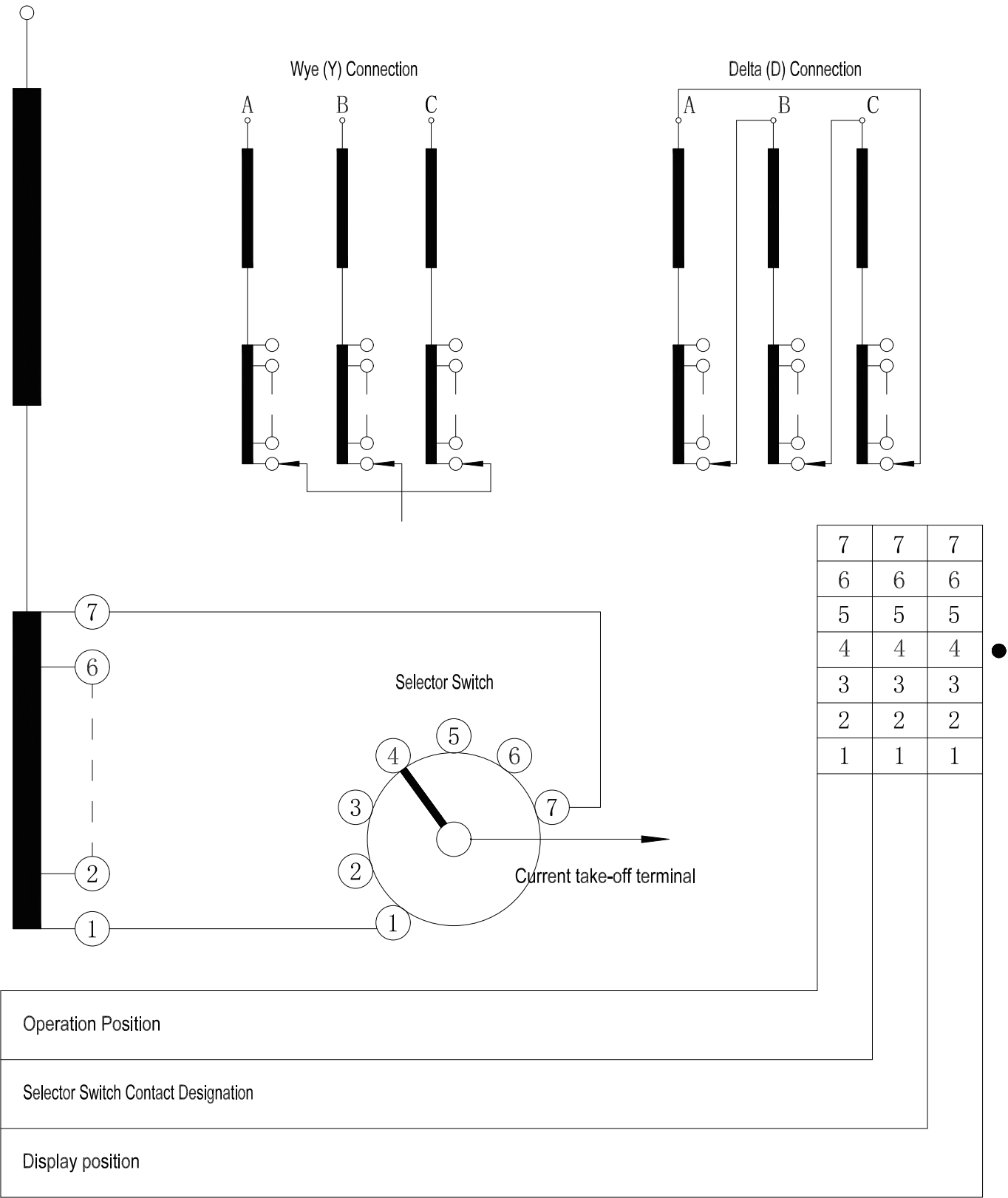
Left Output Shaft Arrangement (with Pressure relief valve/device):



Pressure relief valve/device Electrical Signal Output Wiring Diagram

R: Right Output Shaft
L: Left Output Shaft
P: With Pressure relief valve/device

Figure 14 Basic Connection Diagram of CV2 Type OLTC 10070



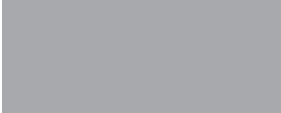


Figure 15 Basic Connection Diagram of CV2 Type OLTC 10071W

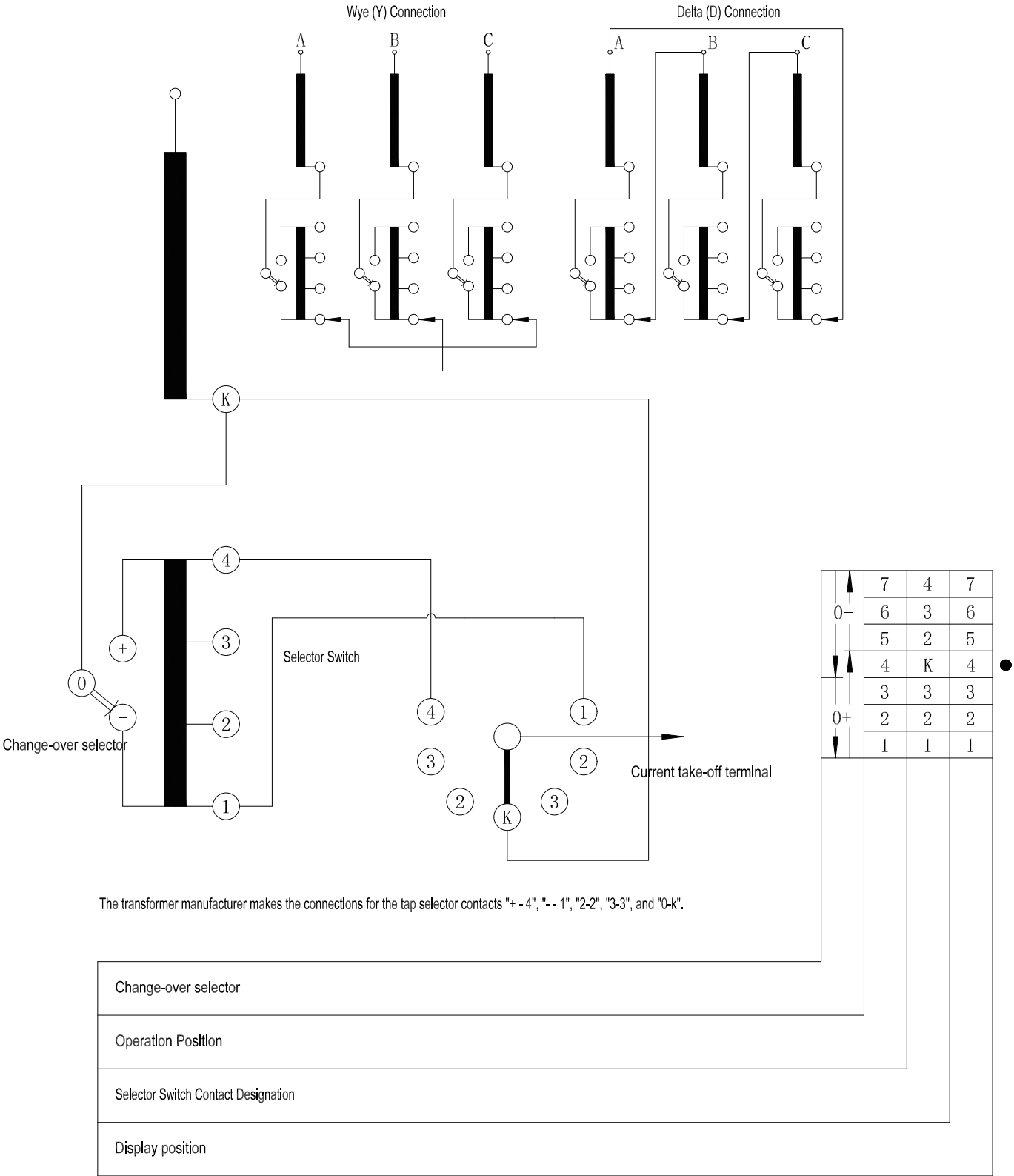
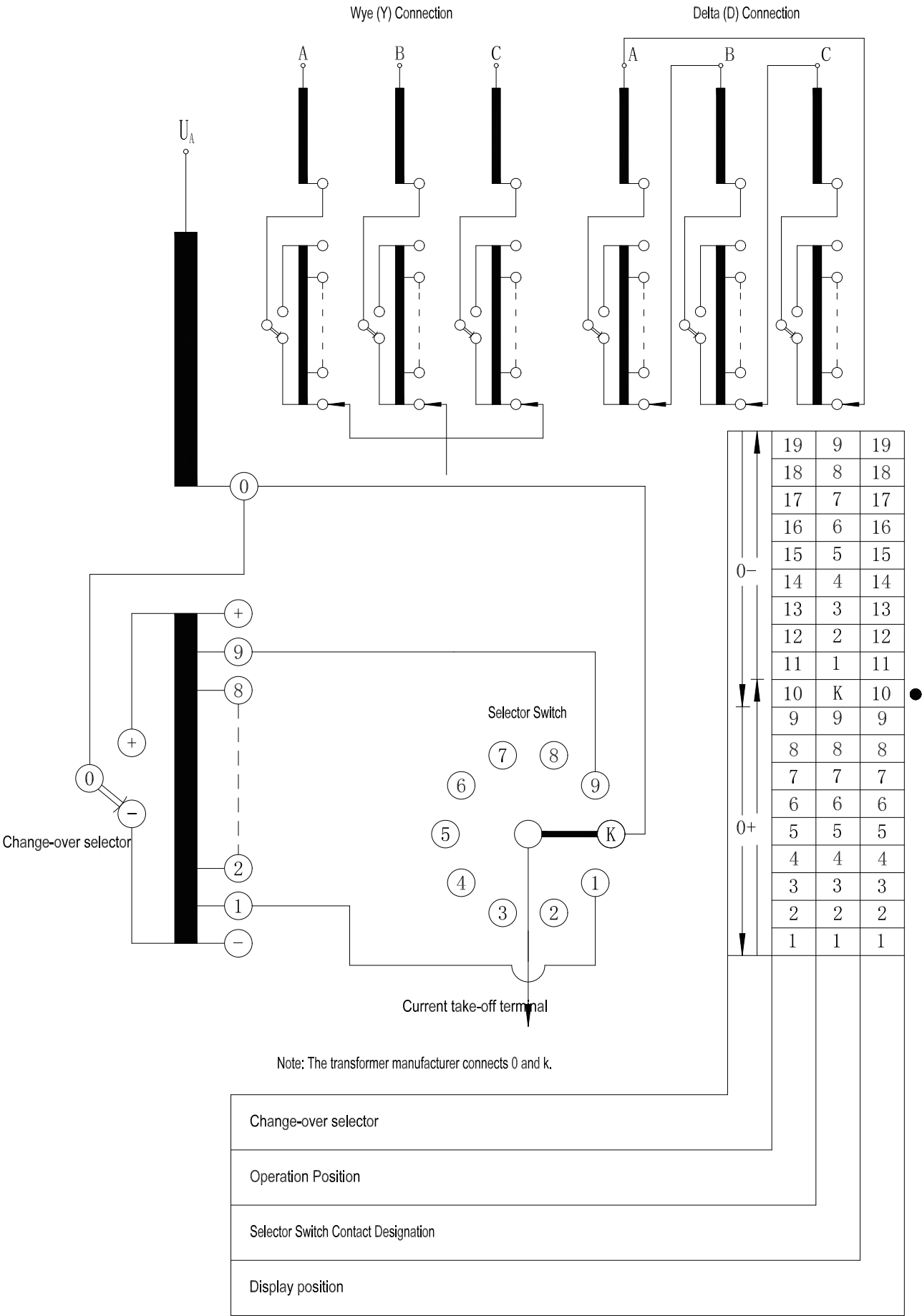


Figure 16 Basic Connection Diagram of CV2 Type OLTC 10191W



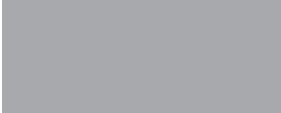


Figure 17 Basic Connection Diagram of CV2 Type OLTC 10191G

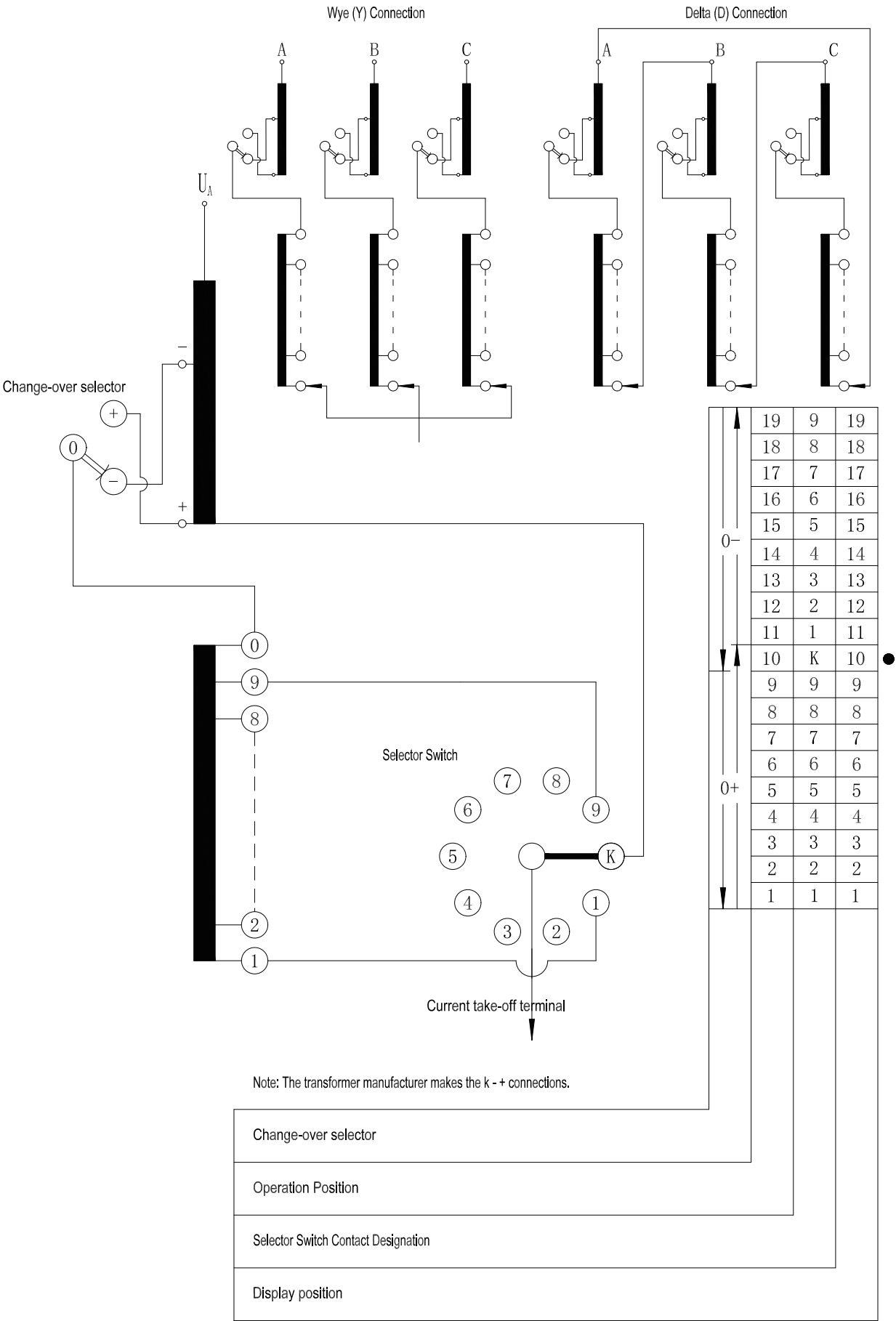
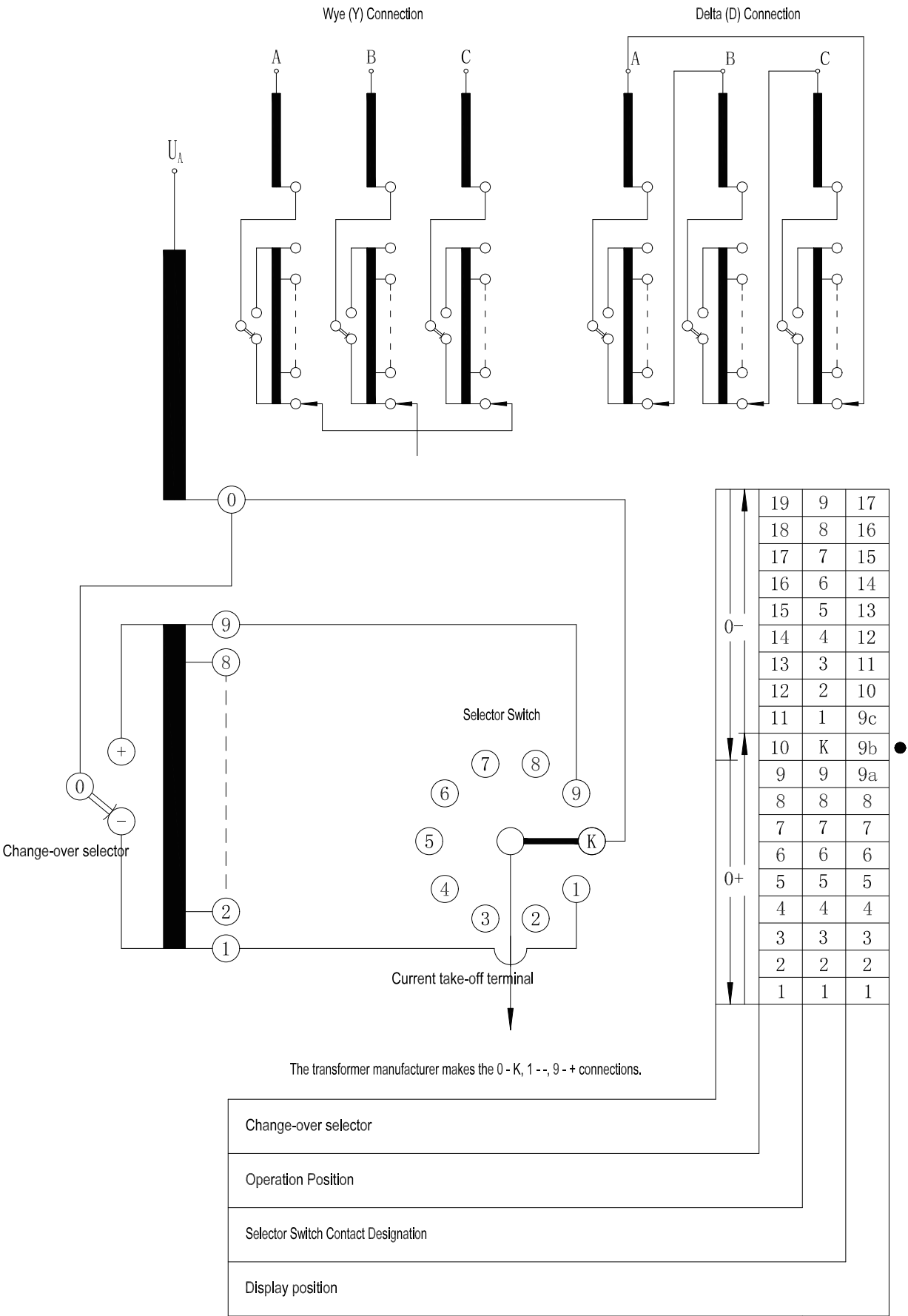


Figure 18 Basic Connection Diagram of CV2 Type OLTC 10193W



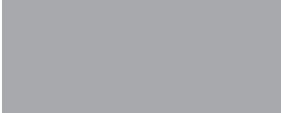


Figure 19 Basic Connection Diagram of CV2 Type OLTC 10193G

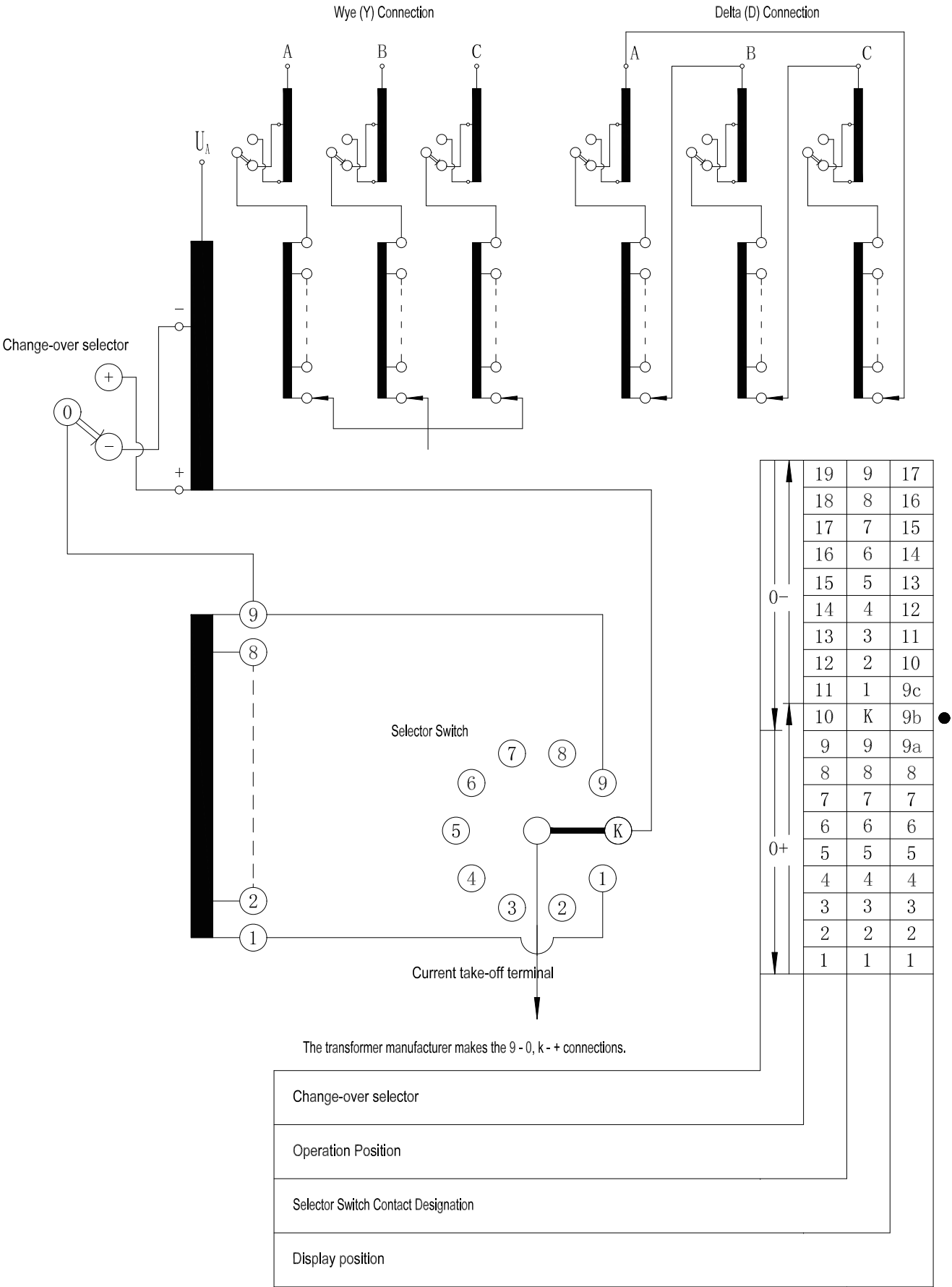
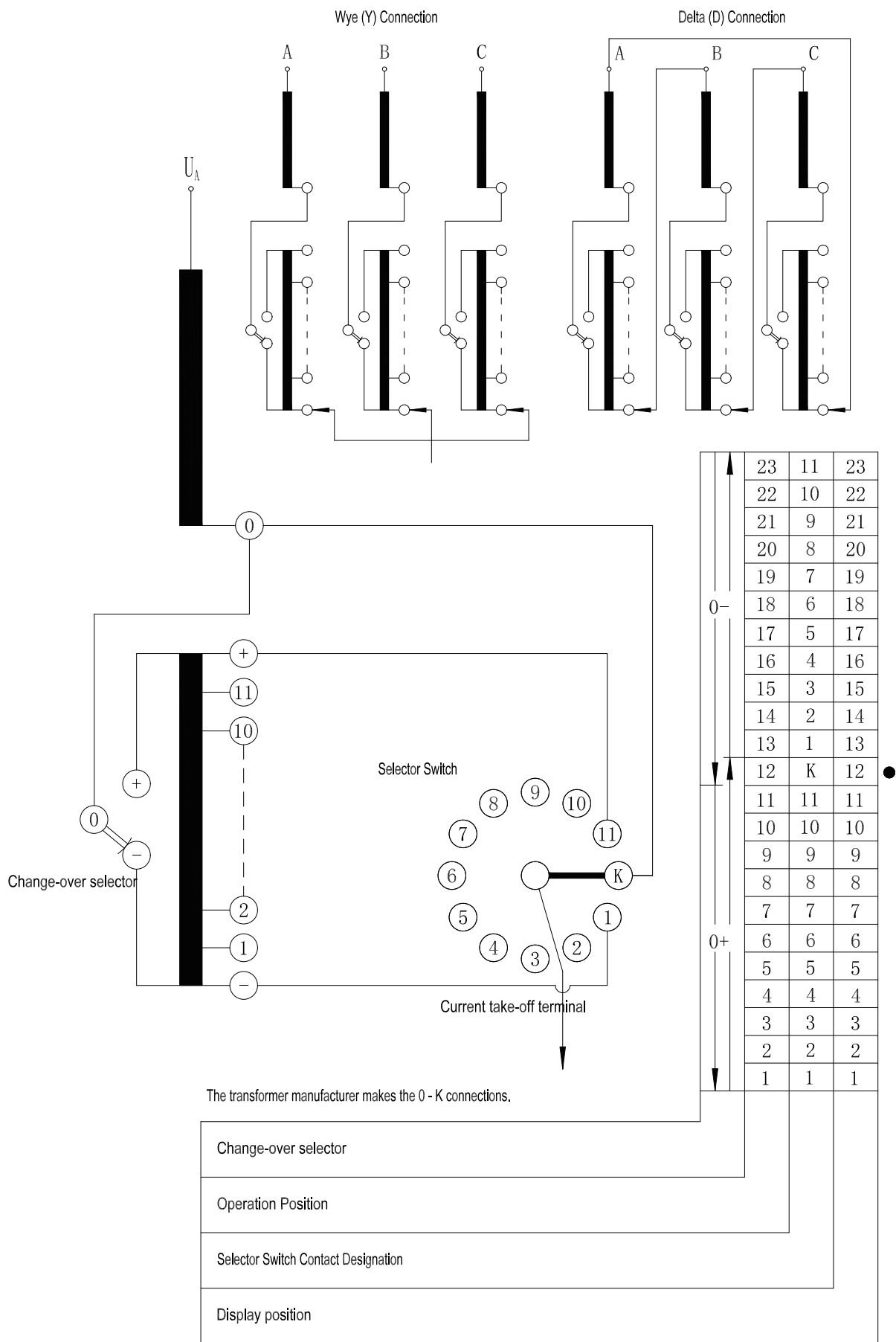


Figure 20 Basic Connection Diagram of CV2 Type OLTC 12231W



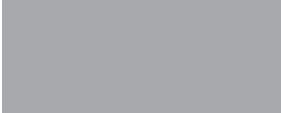


Figure 21 Basic Connection Diagram of CV2 Type OLTC 12231G

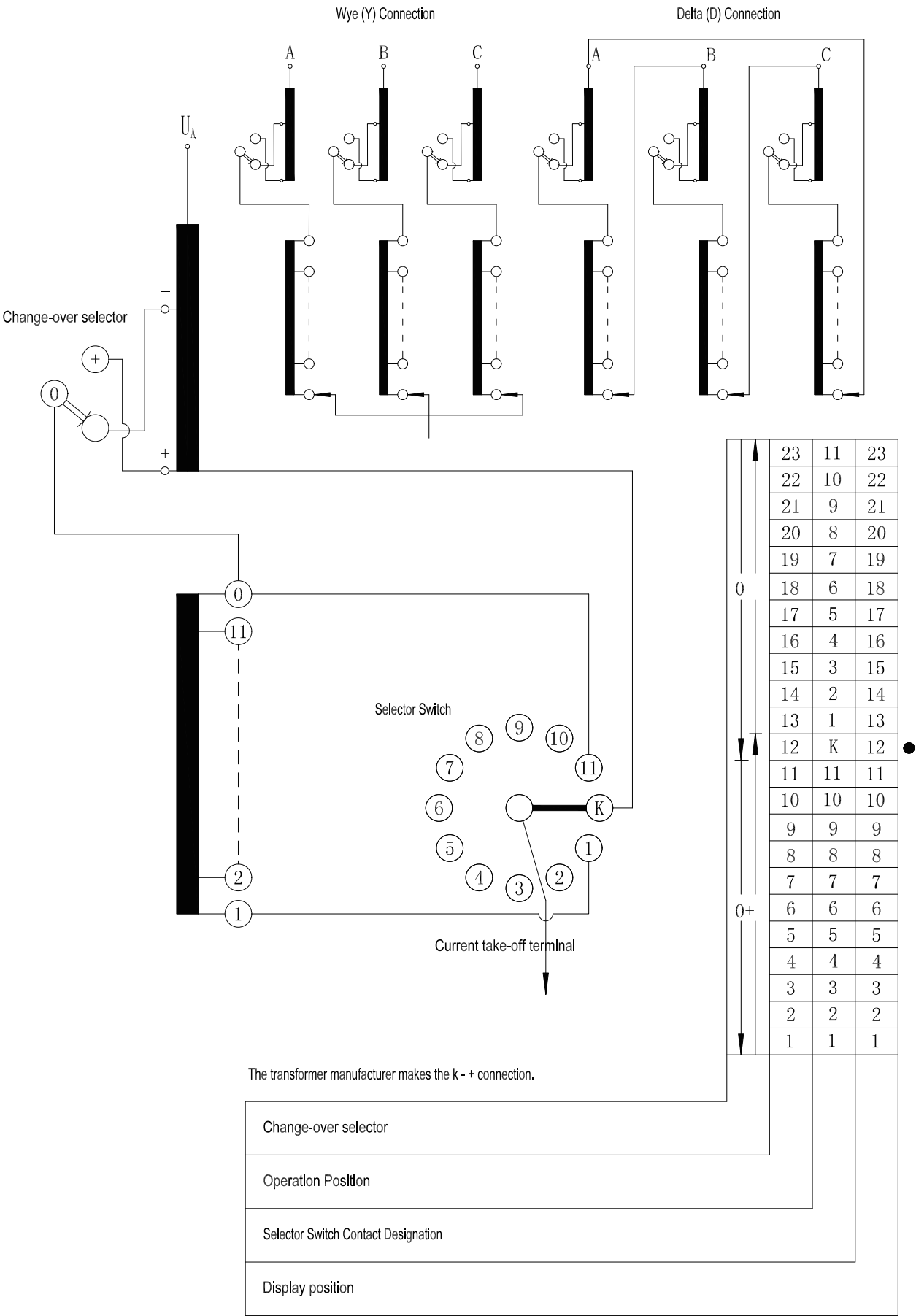
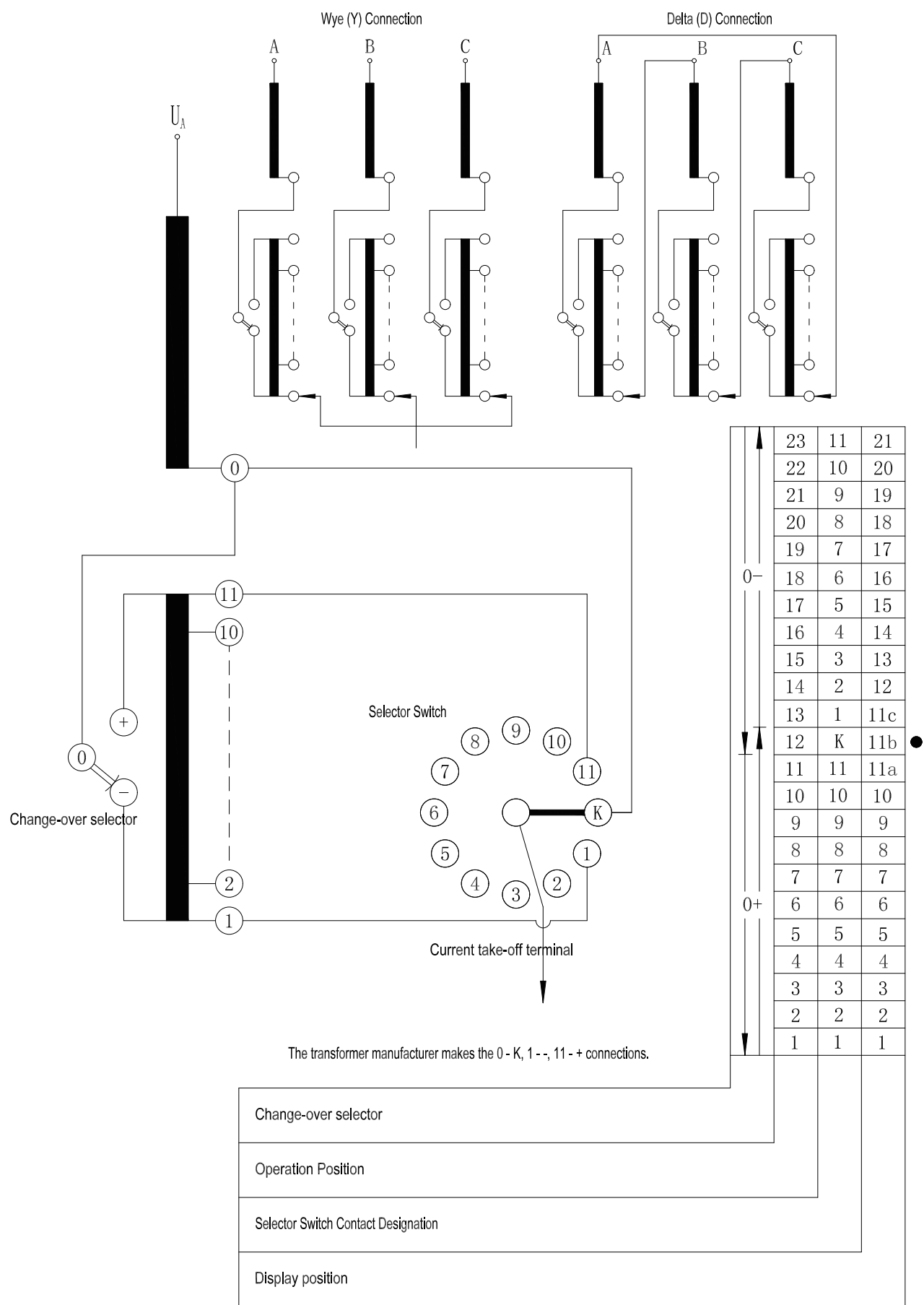


Figure 22 Basic Connection Diagram of CV2 Type OLTC 12233W



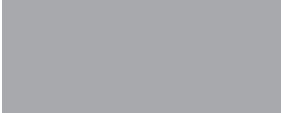


Figure 23 Basic Connection Diagram of CV2 Type OLTC 12233G

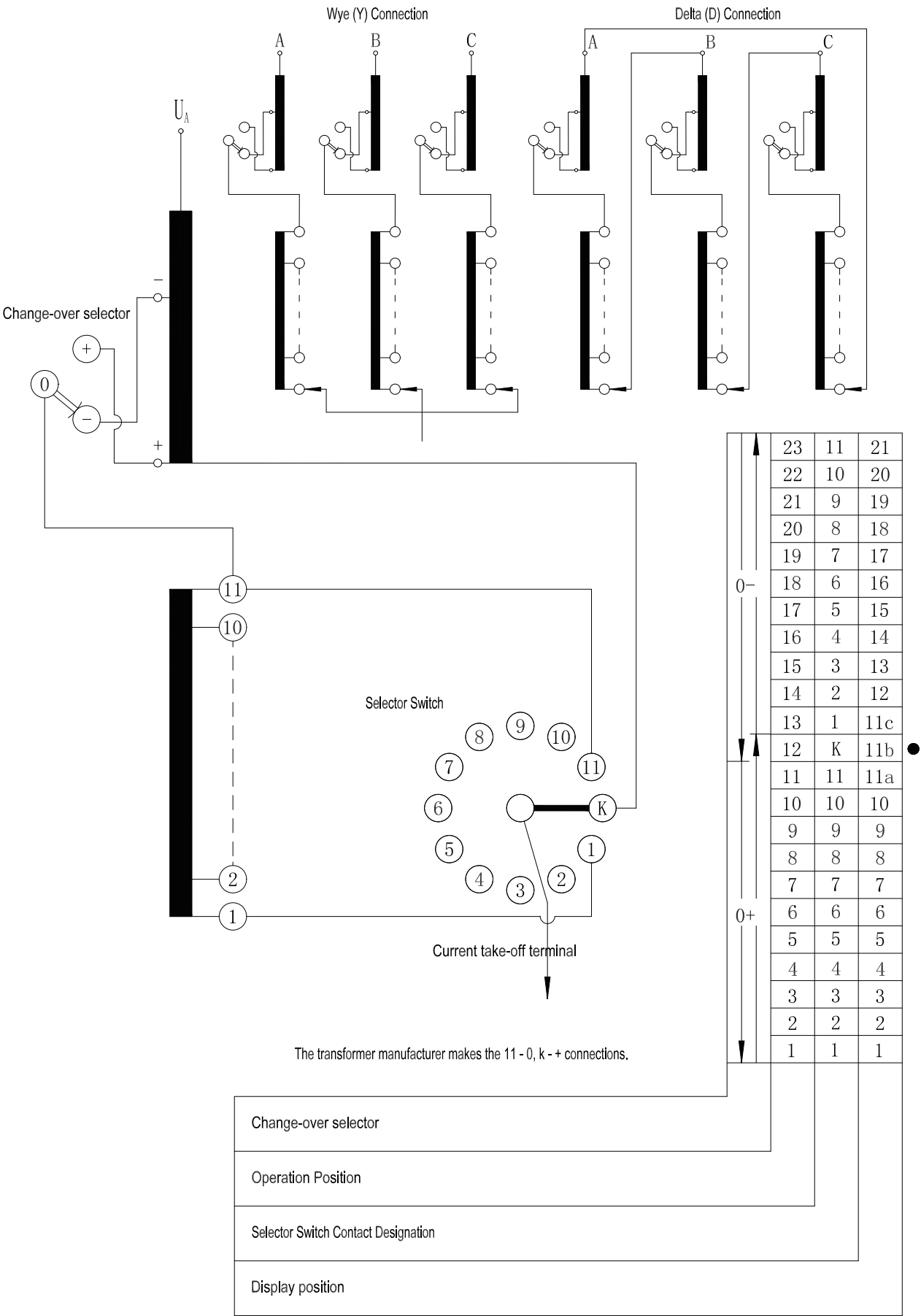
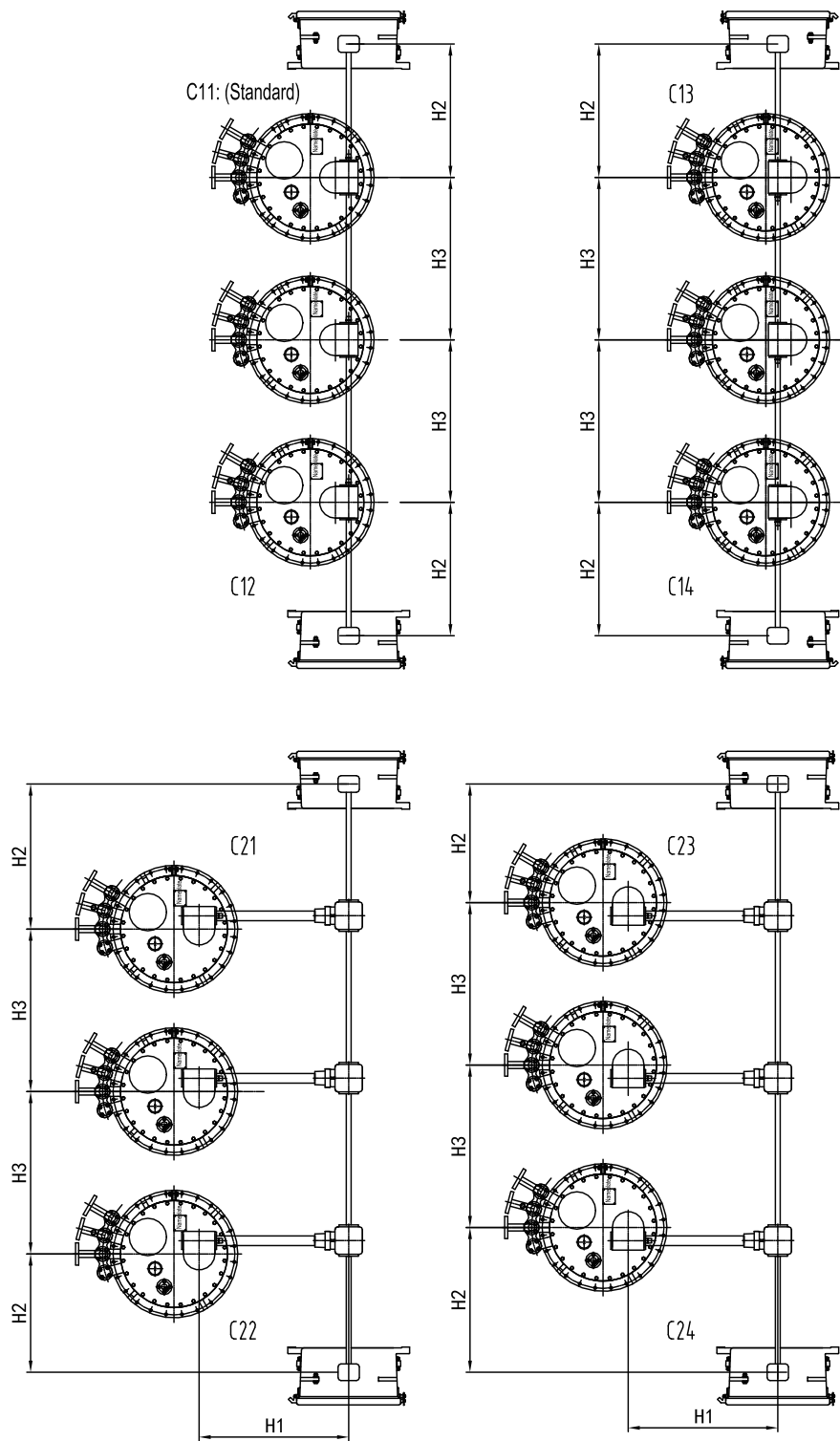


Figure 24 Mechanical Interlocking Arrangement of Three OLTCs

When ordering CV2 three mechanically interlocked switches, select the desired top cover arrangement. Otherwise, it will be produced according to standard drawing C11. For special arrangements, specify requirements at the time of order.



Dimension H	H1 ≥	H2 ≥	H3 ≥
Minimum Dimension (mm)	535	515	840
Mechanical Minimum Dimension			



Figure 25 Outline Dimensions of Bevel Gearbox

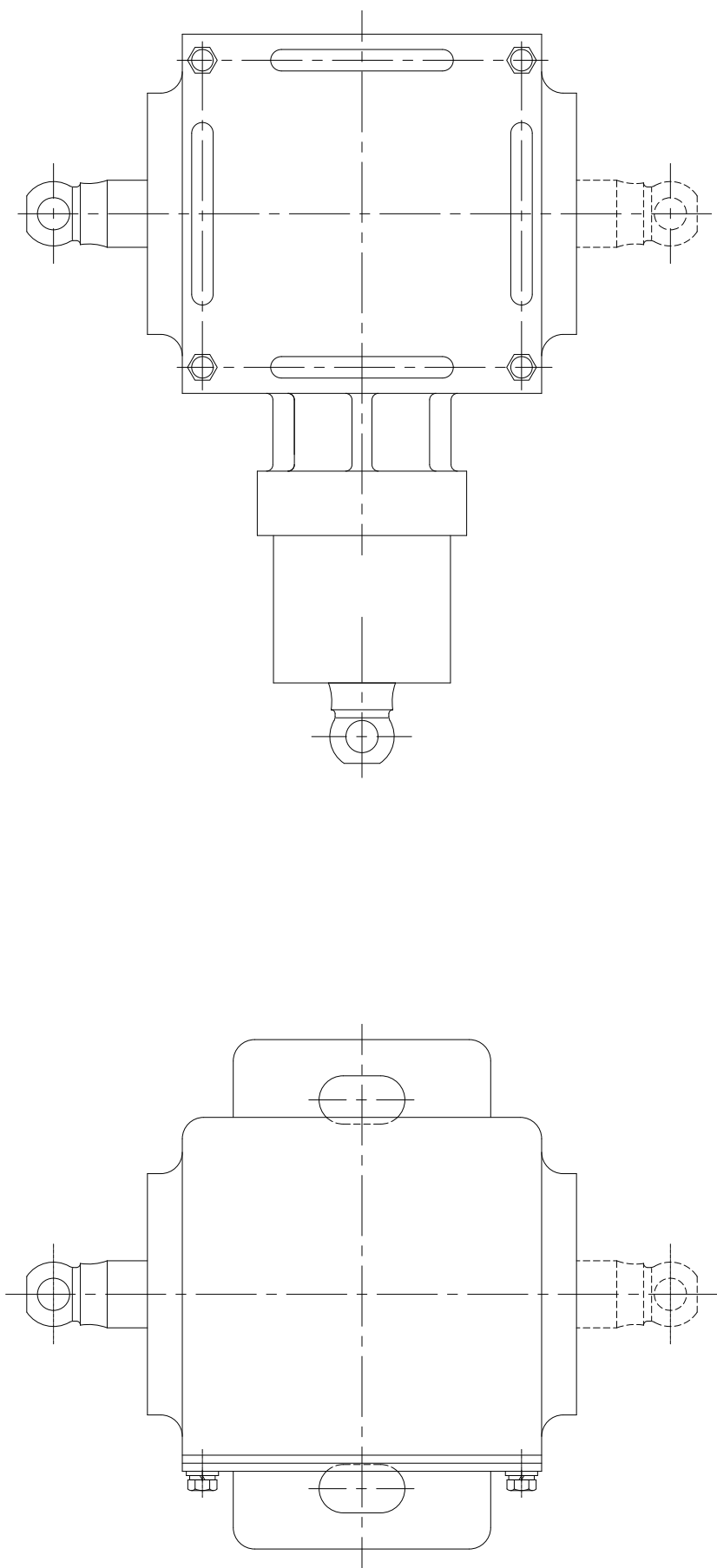
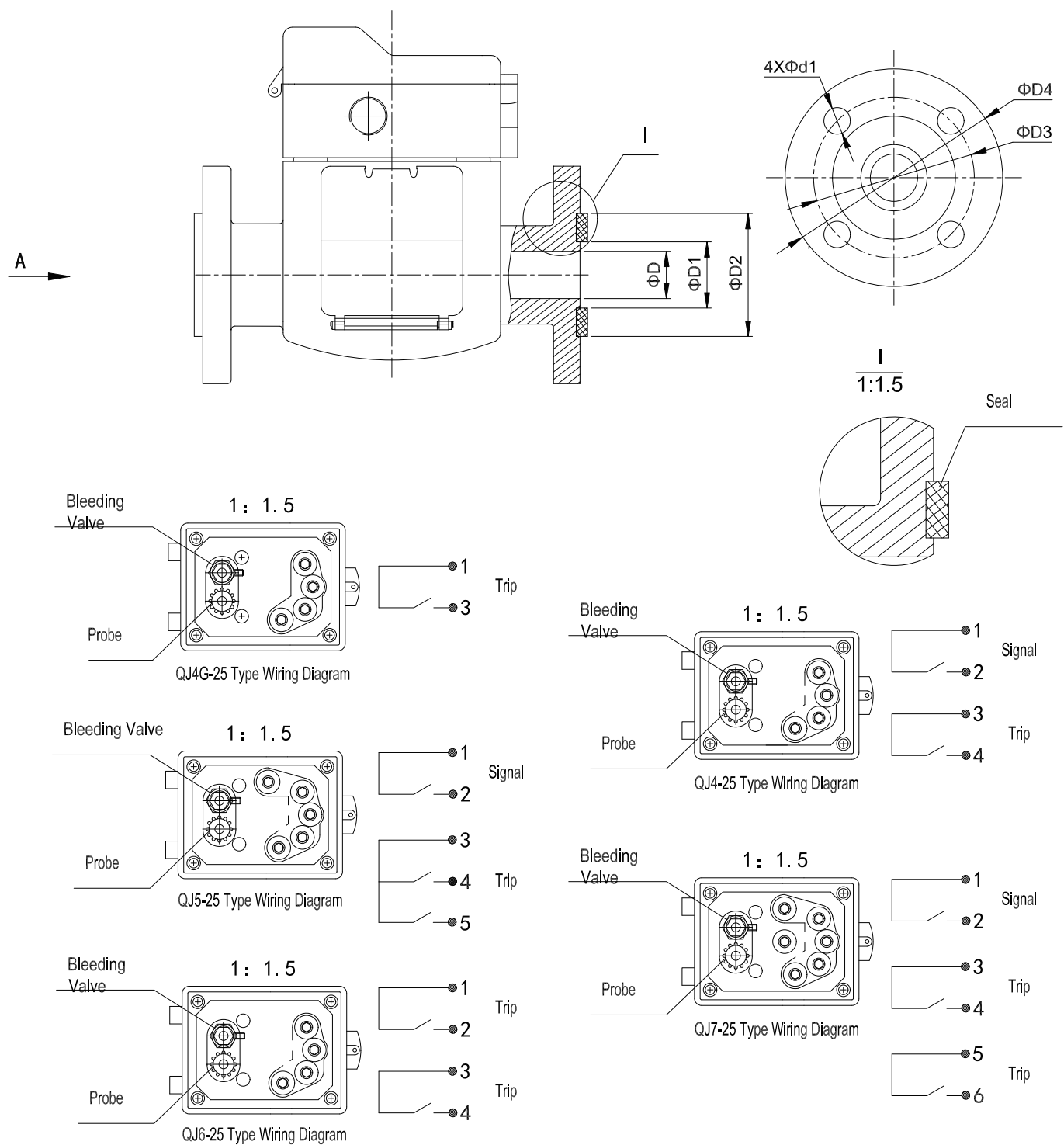


Figure 26 Outline Dimensions of Protective Relay



Model	D	D1	D2	D3	D4	d1	H1	H2	L1	L2	Remarks
QJ4-25	25	35	65	85	115	14	215	158	208	200	Single Signal, Single Trip Coil
QJ4G-25	25	35	65	85	115	14	190	158	208	200	Single-Trip Coil
QJ5-25	25	35	65	85	115	14	215	158	208	200	Single Signal, Double Trip Coil with Common Point
QJ6-25	25	35	65	85	115	14	190	158	208	200	Double Independent Trip Coils
QJ7-25	25	35	65	85	115	14	215	158	208	200	Single Signal, Double Independent Trip Coils



Shanghai Huaming Power Equipment Co., Ltd.
Address: No.977 Tongpu Road, Putuo District, Shanghai, China
Postal Code: 200333
Tel.: +86 21 5270 8966 (Main line)
Fax: +86 21 5270 3385